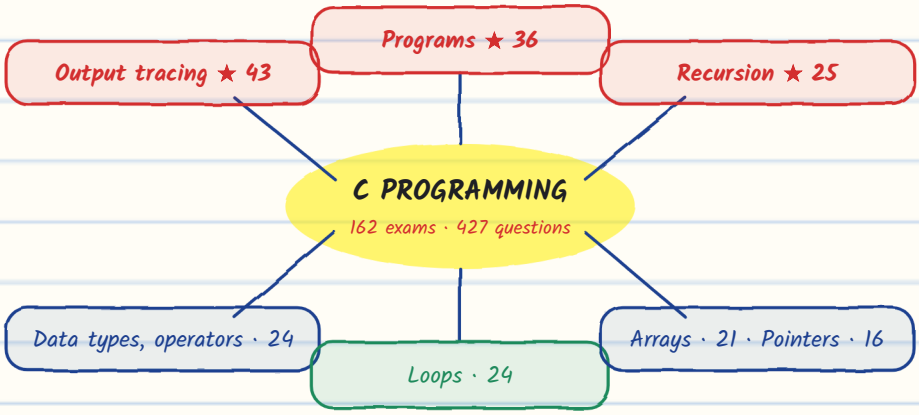
Date: 19.10.2026
topper's notes

Structured Programming (C)

every C / structured-programming question asked in BCS · NTRCA · BPSC · bank · power-sector · BUET-set papers, solved in 41 sheets



↑ the number is how many different exams asked that topic (red = most asked)

➜ How to read a question card

Q The question, as printed in the paper (typos fixed).
Asked in: the exams it was taken from, e.g. BPSC AP 2023 · 45th BCS CSE (2024)

Every program and every "find the output" answer was compiled and run; green = the real output. Where C leaves the result **undefined**, the sheet says so.

Most asked
① call by value vs reference · 5
② recursive factorial · 5
③ compiler vs interpreter · 5 + 3
④ array vs structure · 4
⑤ Fibonacci · local vs global · ++i vs i++ · 2 each

➜ Syllabus coverage

"Structured programming: algorithms, flowcharts, data types, operators, control structures, functions, recursion, arrays, strings, pointers, structures and unions, file I/O", in C.

Legend yellow = key term · red circle = must-memorise · **purple box** = memory trick · **red box** = exam trap · **green** = checked answer

➜ Contents

✓ Languages · translators · errors … p. 2–4
✓ Algorithm, flowchart, GCD … p. 5–7
✓ Structured · data types · operators p. 8–10
✓ Loops · decision programs … p. 11–14
✓ Series · digit · number programs p. 15–20
✓ Functions · call by reference … p. 21–22
✓ Recursion ..... p. 23–24
✓ Arrays · matrices ..... p. 25–28
✓ Pointers ..... p. 29–30
✓ Strings ..... p. 31–32
✓ Structure, union · files … p. 33–35
✓ Find the output I–V ..... p. 36–40
✓ Night-before · practice MCQs … p. 41

1 Programming languages: low, high, "middle"

14 exams

Q1 Difference between high-level and low-level languages, with examples. / How many kinds of programming language are there? Write the advantages and disadvantages of high-level languages. / Write the names of some high-level languages.

✎ **Asked in:** SPCBL SAE 2022 · 15th NTRCA Computer Operation (2019) · SPARRSO TO(IT) 2021

	Machine language	Assembly language	High-level language
written in	0 and 1	mnemonics: `MOV AX, 5`	English-like: `x = a + b;`
translator	**none** (runs directly)	assembler	compiler / interpreter
speed	fastest	fast	slower (after translation)
portable?	no: one CPU	no	yes

Generations: 1GL machine, 2GL assembly, 3GL procedural (C, Pascal), 4GL (SQL), 5GL (Prolog). High-level **pros**: easy, portable, fewer errors, libraries. **Cons**: needs translation, slower, little hardware control. Names: **FORTRAN, COBOL, C, C++, Java, Python, BASIC**.

MCQ as asked	answer	asked in
Which is a high-level programming language?	**C / C++ / Java**	BB AD(General) 2022
Which is high-level? A) Machine code B) C++ C) Assembly D) C	**B and D**	BB AD(General) 2026
Understood without translating / needs no translation	**machine language**	BPSC AP 2022 · Combined Bank Officer(IT) 2020
The first high-level programming language	**FORTRAN** (1957)	Combined Bank Officer(IT) 2020
A programming language requires users to follow rules of ___	**syntax**	Combined Bank Officer(IT) 2020
Object code is the ___ language file that tells the CPU what to do	**machine**	Combined Bank Officer(IT) 2020
___ is natural-language statements that look like programming code	**pseudocode**	Combined Bank Officer(IT) 2020
Making object code of one system work on another	**porting**	Combined Bank Officer(IT) 2020
What kind of language is C? low / mid / high level / none	**see trap**	DoICT AP 2020

2 Compiler vs interpreter, linker, loader, JIT

12 exams · ★ 8 ask the difference

Q2 Distinguish between compiler and interpreter, with examples. / Write the differences between compiler and interpreter.

✎ **Asked in:** 40th BCS (2020) · BCC Programmer 2019 · BPSC ADA 2022 · BPSC AP 2022 · PGCB SAE 2021 · 15th NTRCA Computer Operation (2019) · 16th NTRCA CSE (2019) · SPARRSO TO(IT) 2021 · Combined Bank AP(SB & Janata) 2026 · ICB AMCL AP 2024

Point	Compiler	Interpreter
translates	the whole program at once	line by line, while running
output	an object / executable file	no object file
errors	all listed after compiling	stops at the first error
execution speed	fast	slow (translates every run)
memory	more (object code)	less
source needed to run?	no	yes
examples	**C, C++, Pascal** (gcc, Turbo C)	**Python, BASIC, JavaScript**

Q3 What is a translator program? Write its functions / its types.

✎ **Asked in:** 17th NTRCA School (2023) · 17th NTRCA CSE (2023) · 15th NTRCA Computer Operation (2019)

A translator converts source code to machine code; it also detects and reports syntax errors. Types: **assembler** (assembly → machine), **compiler**, **interpreter**.

MCQ as asked	answer	asked in
Converts an entire program into machine language: interpreter / converter / simulator / compiler	**compiler**	BPSC AP 2022
A ___ translates a file of source code into machine language	**compiler**	Combined Bank Officer(IT) 2020
The system software that creates the final executable file	**linker**	Teletalk Executive 2019

Q4 What is the LEGv8 architecture? An 8086 has code segment 4000H and instruction pointer 9F20H: find the memory address.

✎ **Asked in:** 45th BCS ICT (2024)

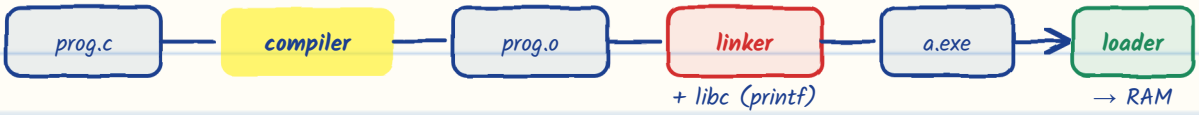
LEGv8 is the teaching subset of ARMv8 (64-bit RISC, 32 registers, load/store) from Patterson & Hennessy. 8086 physical address = segment × 16 + offset = 40000H + 9F20H = **49F20H**.

Exam trap "Is C high- or mid-level?" C is a **high-level** language with low-level features (pointers, bit operators), so Bangladeshi keys usually expect **middle-level**; the source key printed "high level". Write "middle-level (high-level with low-level access)". In BB 2026 both **C++** and **C** are right.

3 From source to running program; kinds of error 8 exams

Q5 What are the tasks of the linker and the loader? Describe with examples. / Write down the role of a linker.

✎ **Asked in:** BPSC AP 2023 · BUET M.Sc. May 2016



Linker: joins the object files and library code into one executable and resolves external references (the call to `printf` is matched to its code in the C library). **Loader**: copies the executable into memory, sets addresses (relocation) and starts it. Errors: "undefined reference to `foo`" is a **linker** error.

- **JIT compiler**: compiles bytecode to native code at run time, method by method, and caches it; the same bytecode runs on any machine that has a JVM, so the program stays machine-independent while running fast. ✎ 46th BCS CSE
- **Comments**: the preprocessor removes every comment, replacing it with one space, before compiling: no code, no size, no speed cost. ✎ BPSC AP 2023
- **Processor per language**: C/C++ compiler; Java compiler + JVM (interpreter + JIT); Python interpreter. ✎ BUET M.Sc. May 2016
- **Platform-independent software** runs unchanged on any OS/CPU, e.g. a Java program (bytecode on any JVM), a Python script, a web app. ✎ Combined Bank SO(IT) 2023

Q6 Explain the errors made during coding, with examples. / Difference between compile-time and run-time errors.

✎ **Asked in:** 16th NTRCA CSE (2019) · RAKUB ANSE 2026

Error	When / who finds it	Example
syntax (compile time)	compiler, before running	`printf("hi")` without `;`
semantic / type	compiler	`int x = "abc";`
linker	linker	`main` missing, undefined function
run-time	while running: program crashes	divide by zero, null pointer
logical	nobody: wrong output	`avg = a + b / 2;`

- Error generated when the rules of a language are violated → **syntax error**. ✎ Teletalk Executive 2019
- Error generated while the program is being executed → **run-time error**. ✎ Combined Bank Officer(IT) 2020

Memory trick The **compiler** finds what you wrote wrong, the **linker** finds what is missing, **run time** finds what goes wrong, and **nobody** finds a logical error — test with sample input.

4 Algorithm, flowchart, program development, errors

11 exams

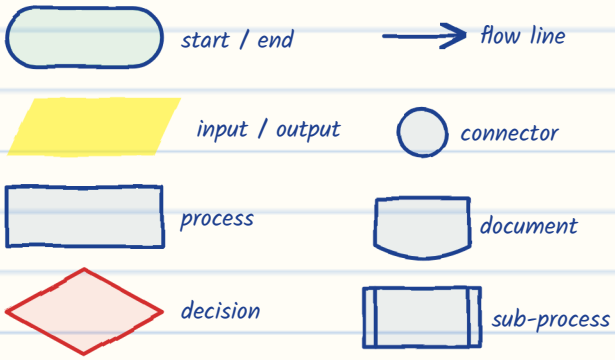
Q7 What is an algorithm? Write its characteristics. / Write the steps of program development. / What is pseudocode?

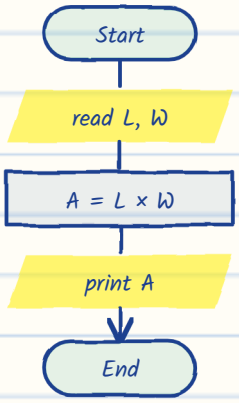
Asked in: 17th NTRCA School (2023) · 15th NTRCA CSE · 17th NTRCA CSE (2023)

An algorithm is a finite, ordered set of unambiguous steps that solves a problem. Characteristics: **input** (zero or more), **output** (at least one), **definiteness**, **finiteness**, **effectiveness**. Pseudocode writes the algorithm in plain English-like statements that look like code. Program development: ① define the problem ② analyse ③ design (algorithm, flowchart) ④ code ⑤ test and debug ⑥ document ⑦ maintain.

Q8 Write the flowchart symbols with their meanings. / Draw a flowchart to find the area of a rectangle.

Asked in: 17th NTRCA School (2023)





Q9 Draw a flowchart to display the numbers 1 to 100.

Asked in: BTCL JAM 2024

Start → i = 1 → **decision i ≤ 100?** → yes: print i, i = i + 1, back to the decision → no: End.

Exam trap The printed chart uses i = i + 2 and "i < 100": that prints only the odd numbers 1–99.

Q10 Draw a flowchart for an ATM PIN check: user input, decision success / failure.

Asked in: Sonali Bank ADA 2026

Start → tries = 0 → read PIN → **PIN correct?** yes → "Success", show menu → End. no → tries = tries + 1 → **tries < 3?** yes → back to read PIN; no → "Failure: card blocked" → End.

Memory trick An algorithm's five properties spell **FIDEO**: **F**initeness, **I**nput, **D**efiniteness, **E**ffectiveness, **O**utput.

5 Largest of three: algorithm, flowchart, program ★ 9 exams

Q11 Write an algorithm, draw a flowchart and write a C program to find the largest of three numbers. / What are algorithm and flowchart? Write an algorithm for the largest of three. / What is pseudocode? Draw a flowchart for the largest of three. / Draw or describe a flowchart to determine the largest of three distinct numbers A, B, C. / Find the maximum of three given numbers.

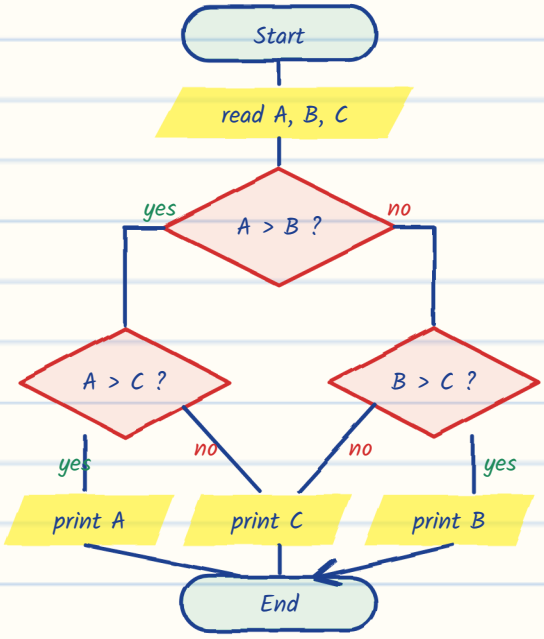
✎ **Asked in:** 13th NTRCA CSE (2016) · 15th NTRCA Computer Operation (2019) · BCPCL JI 2019 · 17th NTRCA CSE (2023) · BTCL JAM 2024 · Combined Bank AP 2026 · Bank (name cut off) AP · DESCO AE(CSE) 2016

Algorithm (pseudocode):

1. Start; read A, B, C.
2. MAX ← A.
3. If B > MAX then MAX ← B.
4. If C > MAX then MAX ← C.
5. Print MAX; stop.

```
int a, b, c, max;
scanf("%d %d %d", &a, &b, &c);
max = a;
if (b > max) max = b;
if (c > max) max = c;
printf("Largest = %d\n", max);
```

input 83 40 3 → Largest = 83



the flowchart the examiners expect

Q12 Write an algorithm and a C program to find the smallest of three numbers. / Flowchart for the smallest of three.

✎ **Asked in:** 18th NTRCA School-2 (2024) · BTCL JAM 2024

```
int a, b, c, min;
scanf("%d %d %d", &a, &b, &c);
min = a;
if (b < min) min = b;
if (c < min) min = c;
printf("Smallest = %d\n", min);
```

Same algorithm and chart with every > turned into <. input 83 40 3 → Smallest = 3

Q13 Write the code for the second highest of three given numbers (83, 40, 3) in C/C++.

✎ **Asked in:** BOF AE(CSE) 2021

```
int a = 83, b = 40, c = 3, mid;
if ((a >= b && a <= c) || (a <= b && a >= c)) mid = a;
else if ((b >= a && b <= c) || (b <= a && b >= c)) mid = b;
else mid = c;
printf("Second highest = %d\n", mid);
```

→ Second highest = 40

Exam trap The printed solution finds the **largest** (83), and its `Int`, `pirintf` do not compile.

6 GCD, LCM and factorial

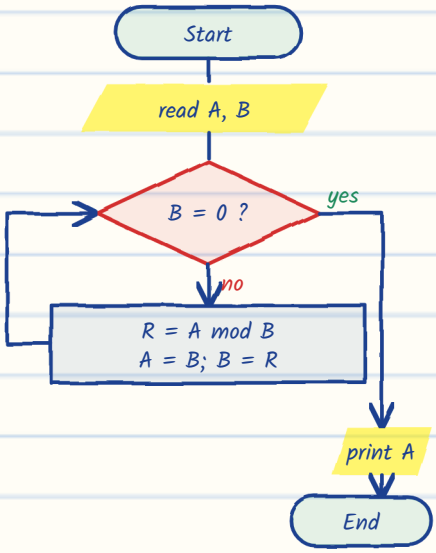
5 exams

Q14 Write an algorithm to find the GCD (গ.সা.গু) of two numbers. / Draw a flowchart and write the algorithm for the GCD of two numbers.

✎ **Asked in:** 17th NTRCA School (2023) · 18th NTRCA School (2024)

Euclid's algorithm: GCD(a, b) = GCD(b, a mod b); stop when b = 0.

1. Start; read A, B.
2. While B ≠ 0: R ← A mod B; A ← B; B ← R.
3. Print A (the GCD); stop.




```
int a, b, r;
scanf("%d %d", &a, &b);
while (b != 0) { r = a % b; a = b; b = r; }
printf("GCD = %d\n", a);
```

input 96 36 → GCD = 12 (96,36 → 36,24 → 24,12 → 12,0)

Q15 Given two integers A and B as input, write a program to compute the least common multiple (LCM) of A and B.

✎ **Asked in:** BB AP 2023

```
long long a, b, x, y, r;
scanf("%lld %lld", &a, &b);
x = a; y = b;
while (y != 0) { r = x % y; x = y; y = r; }   /* x = GCD */
printf("LCM = %lld\n", a / x * b);
```

LCM = a × b / GCD, divide first (a / gcd * b) so the product cannot overflow. input 12 18 → LCM = 36 (the same paper printed it twice, once as "LCN").

Q16 Write an algorithm, draw a flowchart and write a C program to find the factorial of an input number.

✎ **Asked in:** 15th NTRCA CSE

1. Start; read N.
2. F ← 1, I ← 1.
3. While I ≤ N: F ← F × I; I ← I + 1.
4. Print F; stop.

Flowchart: the GCD chart's loop with the decision "I ≤ N ?" and the process box "F = F × I; I = I + 1".

```
int n, i;  long long f = 1;
scanf("%d", &n);
for (i = 1; i <= n; i++) f = f * i;
printf("%d! = %lld\n", n, f);
```

input 5 → 5! = 120

Memory trick GCD loop = "**remainder, shift, repeat**": r = a % b; a = b; b = r. LCM = product ÷ GCD.

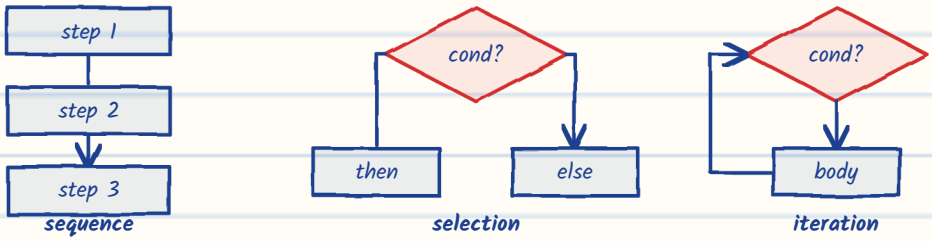
7 Structured programming and problem solving

6 exams

Q17 What is structured programming / a structured programming language? Explain with an example. / What are the main objectives of structured programming? Write two advantages. / Define program, identifier and structured language.

✎ **Asked in:** 38th BCS (2018) · SPARRSO 2020 · 13th NTRCA CSE (2016)

Structured programming writing a program from **three control structures only** — sequence, selection (if/switch), iteration (loops) — divided top-down into functions (modules) with one entry and one exit each, and no goto. Languages: **C, Pascal, ALGOL, Ada**.



Objectives: readable, reliable, easy to test and maintain code. **Advantages**: ① easy to read and debug, because control flows top to bottom; ② modules can be written, tested and reused separately. **Definitions**: a *program* is a set of instructions that tells the computer how to solve a problem; an *identifier* is the name of a variable, function or array (letters, digits, _; not starting with a digit; not a keyword).

Q18 Define problem-solving technique. Write the differences between structured programming and object-oriented programming. / What is programming? Give 4 examples. Difference between structured and OOP programming.

✎ **Asked in:** 45th BCS CSE (2024) · Sonali Bank ADA 2026

Problem-solving technique: a systematic method — understand the problem, divide it into sub-problems, design an algorithm (and flowchart), code, test, refine, document. **Programming** = writing such instructions in a language: e.g. **C, C++, Java, Python**.

	Structured	Object-oriented
built from	functions	classes and objects
design	top-down	bottom-up
data	global, open to every function	hidden inside objects
reuse	copy functions	inheritance
examples	C, Pascal	C++, Java

Memory trick Structured = "**S-S-I**": Sequence, Selection, Iteration — and nothing else.

8 Data types, identifiers, local vs global

13 exams

Q19 Describe the different data types used in C.

✎ **Asked in:** 18th NTRCA College ICT (2024)

type	size*	range / use	format
char	1	−128 … 127, one character	%c
int	4 †	about ±2.1 × 10^9	%d
short · long long	2 · 8	±32 767 · ±9.2 × 10^{18}	%hd · %lld
float · double	4 · 8	≈ 7 · 15 significant digits	%f · %lf
void	—	"no value"	

* bytes by `sizeof` on a 64-bit compiler; † 2 on 16-bit Turbo C. Groups: **basic** (above), **derived** (array, pointer, function), **user-defined** (struct, union, enum, typedef).

Q20 Can you store the value 32678 in a variable of type `int`? If not, why?

✎ **Asked in:** 15th NTRCA · 15th NTRCA ICT

On a **16-bit** compiler (Turbo C) `int` is 2 bytes: range −32 768 … **32 767**, so 32 678 does **not** fit (it wraps to −32 858). Use `long` or `unsigned int`. On today's 32/64-bit compilers `int` is 4 bytes, so it fits — write both.

- Invalid variable name: Average / **No#of-students** / Xzy / y23z → # and - are not allowed. ✎ DoICT ANE 2019 · DoICT AP 2020
- To retain a variable's value between function calls use **static**. ✎ 49th BCS CSE (2025)
- `int x = 5, y = 10, *p = &x; y = *p; y++;` — x before return: **(d) 5** (only the copy in y changes). ✎ BUET M.Sc. April 2021

Q21 What is a variable? Difference between local and global variables.

✎ **Asked in:** 17th NTRCA CSE (2023) · 17th NTRCA ICT (2023) · 17th NTRCA

Local	Global
inside a function / block	outside all functions
seen in that block	seen everywhere
lives while it runs	whole program
garbage at start	0 at start

```c
#include <stdio.h>
int count = 0;                  /* global */
void hit(void) { int n = 1; count += n; }   /* n: local */
int main(void) {
    hit(); hit();
    printf("count = %d\n", count);
    return 0;
}
```

→ count = 2

A variable is a named memory location whose value can change. **Null vs void**: `NULL` is a pointer value pointing nowhere; `void` is a type meaning "no value" (`void f()`) or "any type" (`void *`). ✎ **Asked in:** BCC AP 2023

- **Type conversion**: implicit (`int` → `double` in `2 + 1.5`) or explicit cast `(int)3.9 = 3`. Side effects: loss of fraction, overflow, sign change (−1 → 4294967295 as unsigned), precision loss. ✎ 13th NTRCA CSE (2016)
- **Function prototype** = declaration before use: `int sum(int, int);`. **Overloaded functions** share a name with different parameters: C++ only, not C. ✎ 15th NTRCA CSE

9 Increment, sizeof, precedence, ternary

10 exams

Q22 What are increment and decrement operators? Difference between ++i and i++ (++a and a++) with examples. / What is nested if?

✎ **Asked in:** 17th NTRCA ICT (2023) · NU AP 2020 · 15th NTRCA · 15th NTRCA ICT · 16th NTRCA CSE (2019)

++i (prefix)	i++ (postfix)
increment first, then use	use first, then increment
b = ++i; b gets the new value	a = i++; a gets the old value

```
int i = 5, a, b;
a = i++;      /* a = 5, i = 6 */
b = ++i;      /* i = 7, b = 7 */
printf("a=%d b=%d i=%d\n", a, b, i);
```

→ a=5 b=7 i=7

Alone on a line (i++; or ++i;) both just add 1. **Nested if** = an if inside another if/else: `if (m >= 40) { if (m >= 80) grade = 'A'; else grade = 'P'; } else grade = 'F';`

Q23 What is the difference between `sizeof c+1` and `sizeof(c+1)`? / between `sizeof 'c'+1` and `sizeof('c'+1)`?

✎ **Asked in:** BPSC AP 2023

```
char c = 'x';
printf("%zu %zu\n", sizeof c + 1, sizeof(c + 1));
printf("%zu %zu\n", sizeof 'c' + 1, sizeof('c' + 1));
```

`sizeof` binds tighter than +: `sizeof c + 1` = (1) + 1 = **2**, while `c + 1` is promoted to `int` → **4**. In C a character constant `'c'` is an `int`: **5** and **4**. → 2 4 · 5 4

MCQ as asked	answer	asked in
Which operator has right-to-left associativity? , / + / ?: / =	= (and ?:)	49th BCS ICT (2025)
x = 2 + 3 * 7 % 4 / 2 - 3; x = ?	−1	BPDB AE(CSE) 2021
(x < 0) ? 10 : 15 with x = 5	15	NWPGCL JAM(ICT) 2022
The format %i is used for the ___ type too	int	BUET IICT April 2021

```
int x = 2 + 3 * 7 % 4 / 2 - 3;
printf("%d\n", x);
```

* / % left to right: 3*7 = 21, 21 % 4 = 1, 1 / 2 = **0** (integer division) → 2 + 0 − 3 = −1. → -1

Exam trap In the associativity MCQ **both** ?: and = are right-to-left (the printed note admits it): the key is =; write "= (also ?:)". Precedence, high → low: `() [] → ++ -- ! (unary) * / % + - << >> < > == != & ^ | && || ?: = ,`

10 if-else, loops, break and continue

14 exams

Q24 Write the format of the if...else statement. / What is looping? Why use it? Give an example. / Basic difference between looping and branching.

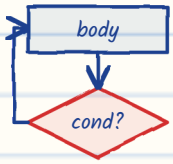
✎ **Asked in:** BPSC ANE 2020 · BCPSC AP 2018

`if (cond) { … } else if (cond2) { … } else { … }`. A loop repeats a block while a condition holds (sum 1 to 100 without 100 lines). **Branching** picks one path once (if, switch); **looping** repeats it.

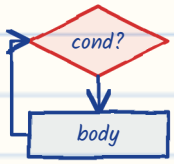
Q25 What is a loop? Write the structure, syntax and flowchart of for, while and do-while loops. / Draw the flowchart of a do-while loop. / Explain the structure of the FOR loop. / Difference between while and do-while.

✎ **Asked in:** 16th NTRCA · 15th NTRCA CSE · 16th NTRCA ICT · Cadet College Lecturer(ICT) 2025 · ICB AMCL AP 2024

loop	syntax
for	`for (init; cond; update) { … }`
while	`while (cond) { … }`
do-while	`do { … } while (cond);`



do-while



while / for

while	do-while
entry-controlled: tests first	exit-controlled: tests after the body
body may run **0** times	body runs **at least once**
no semicolon after `while(…)`	ends with `while(…);`

Q26 Distinguish between break and continue in a loop, with suitable examples.

✎ **Asked in:** Parliament Secretariat SAE 2023

```
for (int i = 1; i <= 6; i++) {
    if (i == 3) continue;   /* skip 3 */
    if (i == 5) break;      /* stop at 5 */
    printf("%d ", i);
}
```

`break` **leaves** the loop (or switch); `continue` **skips the rest of this iteration** and goes to the next. → 1 2 4

MCQ as asked	answer	asked in
Exit-controlled loop? for / while / do while / none	**do while**	49th BCS CSE (2025)
What does `continue` do inside a loop?	**skip to next iteration**	49th BCS CSE (2025)
Keyword to come out of a loop: break / if / switch / continues	**break**	DoICT ANE 2019
`exit()` causes an exit from …	**the program**	BPSC AP 2022

11 Calculator, triangle, bills, tax

9 exams

Q27 Write a program for a basic calculator (+, −, *, /) using a switch statement.

Asked in: 36th BCS (2016)

```
double a, b;  char op;
scanf("%lf %c %lf", &a, &op, &b);
switch (op) {
    case '+': printf("%.2f\n", a + b); break;
    case '-': printf("%.2f\n", a - b); break;
    case '*': printf("%.2f\n", a * b); break;
    case '/': if (b != 0) printf("%.2f\n", a / b);
              else printf("Division by zero\n");
              break;
    default:  printf("Invalid operator\n");
}
```

input 8 * 5 → 40.00

Every `case` ends in **break**, or control falls through to the next case; `default` catches a wrong operator.

Q28 Write a program to find the area of a triangle using IF-ELSE.

Asked in: BCPCL JI 2019

```
double a, b, c, s;
scanf("%lf %lf %lf", &a, &b, &c);
if (a + b > c && b + c > a && a + c > b) {
    s = (a + b + c) / 2;                    /* Heron */
    printf("Area = %.2f\n", sqrt(s * (s - a) * (s - b) * (s - c)));
} else
    printf("Not a triangle\n");
```

input 3 4 5 → Area = 6.00

Q29 Write a Java method to compute an electricity bill: 4.00 per unit up to 100 units, 5.50 for units 101–200, 6.00 for the remaining units.

Asked in: BPDB AE(CSE) 2021

```
public class Main {
    static double bill(int u) {
        if (u <= 100) return u * 4.00;
        if (u <= 200) return 100 * 4.00 + (u - 100) * 5.50;
        return 100 * 4.00 + 100 * 5.50 + (u - 200) * 6.00;
    }
    public static void main(String[] a) { System.out.println(bill(250)); }
}
```

bill(250) → 1250.0 (400 + 550 + 300)

Q30 An employee's weekly pay = hourly wage × regular hours + overtime hours × 1.5 × hourly wage. Read the wage, regular hours and overtime hours and print the total. / Read a cricket player's runs, innings and not-outs and display his average.

Asked in: Combined Bank Officer(IT) 2020 · JBL ASA 2021

```
double w, reg, ot;
scanf("%lf %lf %lf", &w, &reg, &ot);
printf("Pay = %.2f\n", w * reg + ot * 1.5 * w);
```

```
java.util.Scanner sc = new java.util.Scanner(System.in);
int runs = sc.nextInt(), inn = sc.nextInt(), no = sc.nextInt();
System.out.println("Average = " + (double) runs / (inn - no));
```

200 40 5 → Pay = 9500.00 · 450 runs, 12 innings, 2 not out → Average = 45.0 (batting average divides by the times **out**)

12 Slab tax and a database login

4 exams

Q31 Tax slabs: 0–2,50,000 → 0 %; 2,50,001–5,00,000 → 10 %; 5,00,001–10,00,000 → 20 %; above 10,00,000 → 30 %. (a) Write a program to calculate an employee's total tax. (b) From three employees' salaries find the highest tax payer.

✎ **Asked in:** NWPGCL AM(ICT) 2024

```c
/* double tax(double s): slab by slab, top slab first (see note) */
double s[3], best = -1;  int who = 0;
for (int i = 0; i < 3; i++) {
    scanf("%lf", &s[i]);
    printf("Employee %d: tax %.0f\n", i + 1, tax(s[i]));
    if (tax(s[i]) > best) { best = tax(s[i]); who = i + 1; }
}
printf("Highest tax: employee %d\n", who);
```

`tax(): if (s > 1000000) { t += (s-1000000)*0.30; s = 1000000; }` then the same for 5,00,000 at 0.20 and 2,50,000 at 0.10. **Assumption**: the printed slabs are garbled (5000000, 100000); read as the slabs above, taxed slab by slab. → 5000 · 85000 · 185000 · employee 3

Q32 Write a program for a client that uses a database: the client logs in to a page with ID and password. / What is JDBC? Explain the steps to connect a Java application to a MySQL database.

✎ **Asked in:** Combined Bank ADA 2024 · Combined Bank Officer(IT) 2026

JDBC (Java Database Connectivity) is Java's standard API for SQL databases. Steps: ① add the MySQL driver (Connector/J) ② `DriverManager.getConnection(url, user, pass)` ③ create a `PreparedStatement` ④ execute and read the `ResultSet` ⑤ close (try-with-resources).

```c
import java.sql.*;
public class Main {
    static boolean login(String id, String pass) throws SQLException {
        String url = "jdbc:mysql://localhost:3306/bank";
        String sql = "SELECT 1 FROM users WHERE user_id = ? AND password_hash = SHA2(?, 256)";
        try (Connection c = DriverManager.getConnection(url, "app", "secret");
             PreparedStatement ps = c.prepareStatement(sql)) {
            ps.setString(1, id);  ps.setString(2, pass);   /* ? placeholders: no SQL injection */
            try (ResultSet rs = ps.executeQuery()) { return rs.next(); }
        }
    }
}
```

Compiles as shown (it needs a MySQL server to run). Marks: **PreparedStatement** with ? placeholders (never string-concatenate the password into SQL), store a **hash** not the password, close the connection.

13 Printing patterns

4 exams

Q33 Write a C++ program to print the pattern 1 2 3 4 5 / 1 2 3 4 / 1 2 3 / 1 2 / 1. / Find the output: `for (int i = 5; i >= 1; i--) { for (int j = 1; j <= i; j++) printf("%3d", j); }`

✎ **Asked in:** PGCB AE(CSE) 2021 · Cadet College Lecturer(ICT) 2025

```
for (int i = 5; i >= 1; i--) {
    for (int j = 1; j <= i; j++) cout << j << " ";
    cout << "\n";
}
```

1 2 3 4 5 · 1 2 3 4 · 1 2 3 · 1 2 · 1 (one row per line)

Exam trap The Cadet College code has **no newline**: it prints all 15 numbers on one line, each 3 wide: ␣1␣2␣3␣4␣5␣1␣2…␣1.

```
for (int i = 5; i >= 1; i--) { for (int j = 1; j <= i; j++) printf("%3d", j); }
```

→ " 1 2 3 4 5 1 2 3 4 1 2 3 1 2 1" (one line)

Q34 Read an integer and display a pattern: for 1234 print 1234, 234, 34, 4.

✎ **Asked in:** Hydrocarbon Unit AD(ICT) 2025

```
int n, p = 1;
scanf("%d", &n);
while (p <= n) p *= 10;          /* 10000 for 1234 */
for (p /= 10; p >= 1; p /= 10)
    printf("%d\n", n % (p * 10)); /* keep the last digits */
```

input 1234 → 1234 · 234 · 34 · 4

Q35 Print Floyd's (binary) triangle for N: n = 5 → 1 / 01 / 101 / 0101 / 10101.

✎ **Asked in:** Combined Bank AP 2024

```
int n;
scanf("%d", &n);
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= i; j++) printf("%d", (i + j) % 2 == 0);
    printf("\n");
}
```

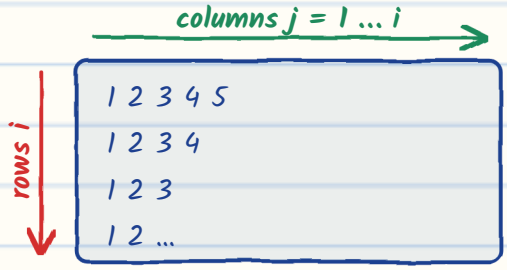
Cell (i, j) is 1 when i + j is even. → 1 · 01 · 101 · 0101 · 10101

Q36 Write a program (or pseudocode) that outputs "EXAM" 100 times.

✎ **Asked in:** Parliament Secretariat Programmer 2023

`for (int i = 1; i <= 100; i++) printf("EXAM\n");` — pseudocode: *FOR i ← 1 TO 100: PRINT "EXAM".*

Memory trick Any pattern = **outer loop for rows, inner loop for columns**. Ask two questions: how many rows (i), and what does row i print (j from … to …)? Put \n after the inner loop.



14 Sum, product, odd, even, squares

9 exams

Q37 Write a C program to find the sum of 1 to N. / Write C programs for the product and the sum of the numbers 1 to 100.

Asked in: 15th NTRCA Computer Operation (2019) · W2PDCL SAE 2023

```
int n;  long long sum = 0;  double prod = 1;
scanf("%d", &n);
for (int i = 1; i <= n; i++) { sum += i; prod *= i; }
printf("sum = %lld, product = %g\n", sum, prod);
```

n = 100 → sum = 5050, product = 9.33262e+157

Exam trap 100! has 158 digits: `int` and `long long` overflow (13! already beats `int`). Use **double** and say so. Check the sum with n(n+1)/2.

Q38 What is a keyword? Write a C program for the sum of all odd numbers 1 + 3 + 5 + … up to 100. / Sum of the even numbers from 1 to 100. / Sum of all even numbers from 1 to n (C or Python).

Asked in: 17th NTRCA CSE (2023) · BCPCL JI 2019 · Combined Bank Officer(IT) 2025

```
int odd = 0, even = 0;
for (int i = 1; i <= 100; i += 2) odd += i;      /* 1,3,...,99 */
for (int i = 2; i <= 100; i += 2) even += i;     /* 2,4,...,100 */
printf("odd = %d, even = %d\n", odd, even);
```

```
n = int(input())
print(sum(range(2, n + 1, 2)))
```

odd = 2500, even = 2550 · Python n = 10 → 30

A **keyword** is a reserved word with a fixed meaning that cannot be an identifier: `int`, `if`, `else`, `for`, `while`, `return`, `void`, `struct` (C has 32 in C89).

Q39 Write a program for $1^2 + 2^2 + 3^2 + \dots + N^2$ using a user-defined function. / Write a C++ program for 1 + (1+2) + (1+2+3) + … + (1+2+…+n).

Asked in: 15th NTRCA CSE · MM AP 2017

```
/* long sumsq(int n): loop s += i*i, return s */
int n;  scanf("%d", &n);
printf("%ld\n", sumsq(n));
```

```
int n;  long term = 0, sum = 0;
cin >> n;
for (int i = 1; i <= n; i++) { term += i; sum += term; }
cout << sum << "\n";
```

N = 5 → 55 (= n(n+1)(2n+1)/6) · n = 4 → 1 + 3 + 6 + 10 = 20

Q40 Find the sum of the series 1 + 2 + 4 + 7 + 11 + … + N (to the n-th term).

Asked in: Combined Bank AP 2024

```
int n, term = 1, sum = 0;
scanf("%d", &n);
for (int i = 1; i <= n; i++) { sum += term; term += i; }  /* gaps 1,2,3,... */
printf("%d\n", sum);
```

Each term adds one more than the last gap. **Assumption**: N = number of terms (the second copy says "n-th term"). n = 5 → 1+2+4+7+11 = 25

Memory trick Every series program is the same **three lines**: sum = 0 · loop over the terms · sum += term. Only "term" changes.

15 e^x, sin x, harmonic and fraction series

7 exams

Q41 Write a program for the series $e^x = 1 + x/1! + x^2/2! + x^3/3! + \dots$

Asked in: Sonali Bank ADBA 2024 · Combined Bank ADA 2024

```
double x, term = 1, sum = 1;  int n;
scanf("%lf %d", &x, &n);              /* x and number of terms */
for (int i = 1; i < n; i++) {
    term = term * x / i;              /* x^i / i! from the last term */
    sum += term;
}
printf("e^%.1f = %.6f\n", x, sum);
```

x = 1, 15 terms → e^1.0 = 2.718282

Build each term from the previous term (x x / i): no pow(), no factorial overflow.

Q42 Write C code for $x - x^3/3! + x^5/5! - x^7/7! + \dots$

Asked in: PGCB AE(CSE) 2024

```
double x, term, sum;  int n;
scanf("%lf %d", &x, &n);
term = sum = x;
for (int i = 1; i < n; i++) {
    term = -term * x * x / ((2 * i) * (2 * i + 1));
    sum += term;
}
printf("sin(%.1f) = %.6f\n", x, sum);
```

This is the Taylor series of sin x (x in radians); each term = previous × (−x²) / ((2i)(2i+1)). x = 1 → sin(1.0) = 0.841471

Q43 Write a program for 1 + 1/2 + 1/3 + … + 1/n, n read from input. / What is an array processor? Write a C program for 1 + 1/2 + 1/3 + … + 1/N.

Asked in: Fisheries & Livestock Instructor(CS) 2023 · 16th NTRCA CSE (2019)

```
int n;  double s = 0;
scanf("%d", &n);
for (int i = 1; i <= n; i++) s += 1.0 / i;
printf("%.6f\n", s);
```

Write 1.0 / i: 1 / i is integer division and gives 0 after the first term. n = 5 → 2.283333

An array processor (vector/SIMD processor) applies one instruction to many data elements in parallel.

Q44 Write a C/C++ program for the series up to n: 1/(2×3) + 2/(3×4) + 3/(4×5) + …

Asked in: NESCO AM(ICT) 2021

```
int n;  double s = 0;
scanf("%d", &n);
for (int i = 1; i <= n; i++) s += (double) i / ((i + 1) * (i + 2));
printf("%.4f\n", s);
```

n = 3 → 1/6 + 2/12 + 3/20 = 0.4833

Q45 Write a program to find the minimum number from a series.

Asked in: BTCL AE(CSE) 2017

```
int n, x, min;
scanf("%d %d", &n, &min);             /* n, then the first number */
for (int i = 1; i < n; i++) { scanf("%d", &x); if (x < min) min = x; }
printf("Minimum = %d\n", min);
```

Start min at the **first** number, not 0 (0 would be wrong for all-positive input). 5 numbers 7 −2 9 0 4 → Minimum = -2

16 Working digit by digit: n % 10 and n / 10

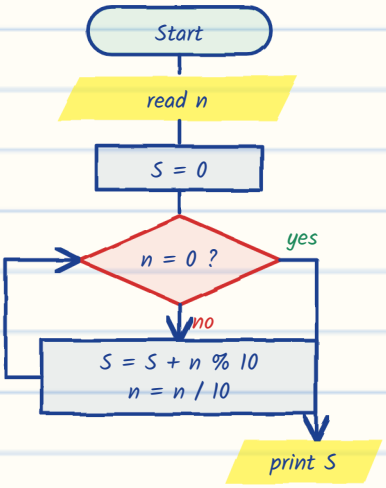
9 exams

Q46 Write a C program to sum all the digits of a 5-digit number. Draw the flowchart of this program. / Write a program in C for the sum of the digits of a number. 4 / Write the function `sumdigits` for the given `main` (input 123 → output 6).

✎ **Asked in:** 45th BCS CSE (2024) · Parliament Secretariat SAE 2023 · BTCL AM(Technical) 2021

```
int sumdigits(int n);              /* defined below */
int n;
scanf("%d", &n);
if (n < 10000 || n > 99999) printf("Not a 5-digit number\n");
else printf("Sum of digits of %d = %d\n", n, sumdigits(n));
/* int sumdigits(int n) { int s = 0;
       while (n != 0) { s += n % 10; n /= 10; }   // last digit, drop it
       return s; }                                          */
```

input 12345 → Sum of digits of 12345 = 15



Q47 Find the sum of the digits of a number until the sum is a single digit (12345 → 6). / Find the summation of a number until it becomes a single digit. / Count the number of digits in a given number.

✎ **Asked in:** Combined Bank SO 2020 · Sonali Bank Officer(IT) 2021 · BKSP AP 2022

```
int n, s, digits = 0, t;
scanf("%d", &n);
for (t = n; t != 0; t /= 10) digits++;      /* count digits */
while (n > 9) {                             /* digital root */
    for (s = 0; n != 0; n /= 10) s += n % 10;
    n = s;
}
printf("digits = %d, single digit = %d\n", digits, n);
```

12345 → 15 → 6. Short cut: digital root = 1 + (n − 1) % 9 for n > 0. → digits = 5, single digit = 6 (input 0 has 1 digit: handle it separately).

Q48 A string contains letters and digits; print the sum of all the digit characters. / Score each letter (A 1, B 2, C 3, D 4, E 5, F 1, G 2, … Z 1) and sum the scores of a word; non-letters score nothing.

✎ **Asked in:** Jamuna Oil JO(IT & MIS) 2025 · DG Food AP 2022

```
char s[100], w[100];  int digits = 0, score = 0;
scanf("%s %s", s, w);
for (int i = 0; s[i]; i++)
    if (isdigit(s[i])) digits += s[i] - '0';      /* '7' - '0' = 7 */
for (int i = 0; w[i]; i++)
    if (w[i] >= 'A' && w[i] <= 'Z') score += (w[i] - 'A') % 5 + 1;
printf("digit sum = %d, score = %d\n", digits, score);
```

Character arithmetic: `c - '0'` turns a digit character into its value; the score repeats every 5 letters, so (position % 5) + 1. a1b2c3x9 → 15 · HELLO → 3+5+2+2+5 = 17

17 Reverse a number, palindrome number

7 exams

Q49 Write a function in a structured language to reverse an integer, with the prototype `int reverse_int(int num)` (4567 → 7654). / Write a C program to reverse an integer. / Find the reverse for one positive and one negative input.

✎ **Asked in:** 46th BCS ICT · BCC AP 2017 · BAERA Engineer(CSE) January 2021

```
/* int reverse_int(int num) {
       int rev = 0;
       while (num != 0) { rev = rev * 10 + num % 10; num /= 10; }
       return rev; }                                          */
printf("%d %d %d\n", reverse_int(4567), reverse_int(-123), reverse_int(220));
```

→ 7654 -321 22

In C, `-123 % 10` = −3 and `-123 / 10` = −12 (truncation toward zero), so a negative number reverses with its sign: **−321**. (The prototype's `reverse-int` is not a legal name; use `reverse_int`.)

Q50 Find the output for input 6789 (and for any input): `while (number != 0) { remainder = number % 10; rev = rev * 10 + remainder; number /= 10; } printf("%d", rev);` / the same as a function `reverseNumber(num)`.

✎ **Asked in:** BGDCL AM(CSE) 2024

pass	number	remainder	rev = rev × 10 + rem
1	6789	9	9
2	678	8	98
3	67	7	987
4	6	6	**9876**

→ Reversed Number: 9876 · the function version: Input: 6789 / Output: 9876

Q51 Write a C/C++/Java program to reverse the digits of a number. Sample input 220, sample output 022.

✎ **Asked in:** JBL ANE 2022

```
int n;
scanf("%d", &n);
do { printf("%d", n % 10); n /= 10; } while (n != 0);
printf("\n");
```

The sample wants **022**: an `int` cannot hold a leading 0, so print each digit instead of building a number. 220 → 022

Q52 Check whether a given integer is a palindrome or not.

✎ **Asked in:** BUET AP(Exam Controller) 2025

```
int n, t, rev = 0;
scanf("%d", &n);
for (t = n; t > 0; t /= 10) rev = rev * 10 + t % 10;
printf("%d is %sa palindrome\n", n, (n == rev) ? "" : "not ");
```

A palindrome reads the same reversed: compare n with its reverse (keep the original in n, reverse a copy t). 12321 → is a palindrome; 1231 → not.

18 Prime, Armstrong, binary, case, quadratic

10 exams

Q53 Write an algorithm / a C program to check whether a number is prime. / Show the prime numbers from 1 to n (n from user input; sample n = 20 → Prime Numbers: 2, 3, 5, 7, 11, 13, 17, 19,).

✎ **Asked in:** NSDA AP 2022 · Parliament Secretariat SAE 2023 · ICB AMCL AP 2024 · Combined Bank AP 2024 · Combined Bank SO(IT) 2024

```
/* int isPrime(int n) { if (n < 2) return 0;
      for (int i = 2; i * i <= n; i++) if (n % i == 0) return 0;
      return 1; }                                              */
int n;
printf("Enter value of n: ");  scanf("%d", &n);
printf("Prime Numbers: ");
for (int i = 2; i <= n; i++) if (isPrime(i)) printf("%d, ", i);
```

→ Enter value of n: Prime Numbers: 2, 3, 5, 7, 11, 13, 17, 19,

Algorithm: read n; if n < 2 → not prime; for i = 2 while i × i ≤ n: if n mod i = 0 → not prime, stop; otherwise prime. Testing up to √n is enough (the printed NSDA answer goes to n/2: correct but slower).

Q54 Write a program to check whether a number is Armstrong or not. / Binary to decimal conversion program (C/C++). / Write a C program for upper-case to lower-case conversion.

✎ **Asked in:** Combined Bank SO(IT) 2020 · DUET AP 2025 · DMTCL AE(ICT) 2023

```
int n, t, d = 0, sum = 0, dec = 0;  char bin[40], s[50];
scanf("%d %s %s", &n, bin, s);
for (t = n; t; t /= 10) d++;                          /* number of digits */
for (t = n; t; t /= 10) sum += (int) pow(t % 10, d);      /* 1^3 + 5^3 + 3^3 */
printf("%d is %sArmstrong\n", n, sum == n ? "" : "not ");
for (int i = 0; bin[i]; i++) dec = dec * 2 + (bin[i] - '0');   /* 1011 → 11 */
printf("binary %s = %d\n", bin, dec);
for (int i = 0; s[i]; i++) if (s[i] >= 'A' && s[i] <= 'Z') s[i] += 32;   /* 'A' 65 → 'a' 97 */
printf("%s\n", s);
```

→ 153 is Armstrong · binary 1011 = 11 · hello

Armstrong number: sum of each digit raised to the number of digits equals the number (153 = 1 + 125 + 27; also 370, 371, 407). Binary → decimal: **double and add** each bit (Horner). Upper → lower: add 32, or `tolower()`.

number	test in one line	examples
prime	no divisor from 2 to √n	2, 3, 5, 7, 11
Armstrong	Σ digit^k = n (k = number of digits)	153, 370, 371, 407
perfect	sum of proper divisors = n	6, 28, 496
palindrome	n = reverse(n)	121, 12321
strong	Σ digit! = n	145 = 1! + 4! + 5!

19 $ax^2 + bx + c = 0$: flowchart and program

2 exams

Q55 Draw the flowchart and write C code to find the roots (value of x) of $ax^2 + bx + c = 0$.

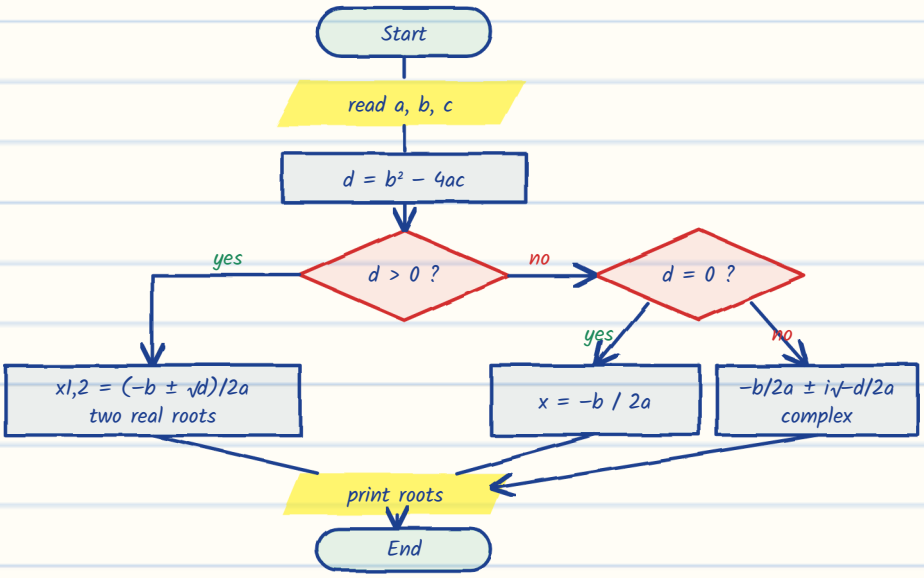
Asked in: 16th NTRCA · 16th NTRCA ICT

```
double a, b, c, d;
scanf("%lf %lf %lf", &a, &b, &c);
d = b * b - 4 * a * c;
if (d > 0)       printf("x1 = %.2f, x2 = %.2f\n", (-b + sqrt(d)) / (2 * a), (-b - sqrt(d)) / (2 * a));
else if (d == 0) printf("x = %.2f\n", -b / (2 * a));
else             printf("x = %.2f ± %.2fi\n", -b / (2 * a), sqrt(-d) / (2 * a));
```

1 −3 2 → x1 = 2.00, x2 = 1.00 · 1 2 1 → x = -1.00 · 1 2 5 → x = -1.00 ± 2.00i

The discriminant $d = b^2 - 4ac$ decides: $d > 0$ two real roots, $d = 0$ one, $d < 0$ complex. Check $a \neq 0$ first (else it is linear).

a, b, c	$d = b^2 - 4ac$	case	roots (program output)
1, −3, 2	9 − 8 = 1	$d > 0$	x1 = 2.00, x2 = 1.00
1, 2, 1	4 − 4 = 0	$d = 0$	x = −1.00 (one repeated root)
1, 2, 5	4 − 20 = −16	$d < 0$	x = −1.00 ± 2.00i



the flowchart NTRCA asks for, with the three cases of d

Memory trick **% 10 peels, / 10 drops, × 10 + pushes** (digit programs) · **d decides** (quadratic): plus → 2, zero → 1, minus → complex.

20 Parameter passing: value vs reference

★ most asked · 11 exams

Q56 Differences between "call by value" and "call by reference". / What is parameter passing? Explain with examples. / Illustrate the difference in a function with examples.

✎ **Asked in:** 17th NTRCA ICT (2023) · BPSC ADA 2022 · BPSC AP 2022 · BPSC AP 2023 · PSB AP 2018 · 15th NTRCA · 15th NTRCA ICT · 13th NTRCA CSE (2016) · 46th BCS ICT

Point	Call by value	Call by reference
what is passed	a copy of the value	the address of the variable
change inside the function	does **not** affect the caller	changes the caller's variable
memory	a new copy	same location, no copy
C syntax	`void f(int x)` · `f(a)`	`void f(int *x)` · `f(&a)`
safety / speed	safe; slow for big data	fast; can change data by mistake

Q57 Write a simple program in C to swap two integers using call by value. / Mention two basic differences between call by value and call by reference; write a program to swap two integers using call by reference.

✎ **Asked in:** BPSC AP 2023 · 46th BCS CSE

```c
#include <stdio.h>
void swapv(int x, int y) { int t = x; x = y; y = t;
    printf("inside swapv: %d %d\n", x, y); }
void swapr(int *x, int *y) { int t = *x; *x = *y; *y = t; }
int main(void) {
    int a = 10, b = 20;
    swapv(a, b);  printf("after swapv: %d %d\n", a, b);
    swapr(&a, &b); printf("after swapr: %d %d\n", a, b);
    return 0;
}
```

→ inside swapv: 20 10
→ after swapv: 10 20
→ after swapr: 20 10

Exam trap "Swap using call by value" can only swap the **copies**: print inside the function to show it worked there and that the caller is unchanged.

Q58 Write a C code segment / program to swap two numbers without using a third variable.

✎ **Asked in:** W2PGCL AE(CSE) 2023 · 45th BCS ICT (2024)

```c
int a = 5, b = 9;
a = a + b;   /* 14 */
b = a - b;   /* 5  */
a = a - b;   /* 9  */
printf("a=%d b=%d\n", a, b);
```

Or with XOR: `a ^= b; b ^= a; a ^= b;` (no overflow risk). → a=9 b=5

- **Call by name** (ALGOL): the argument expression is re-evaluated each time the parameter is used; call by value copies once; call by reference passes the address. Read the scenario: "change seen by the caller" → reference; "copy" → value; "expression evaluated again" → name. ✎ BAERA Engineer(CSE) November 2021
- `typedef struct { int roll; char name[28]; double cgpa; } S;` passed by value copies the **whole struct**, array included: 4 + 28 + 8 = **(b) 40** bytes ("arrays by reference" applies only to bare arrays). ✎ BUET M.Sc. April 2021

21 Functions: syntax, library vs user-defined

6 exams

Q59 Write about the syntax of a function.

Asked in: BARC Programmer 2023

```c
#include <stdio.h>
int add(int a, int b);          /* prototype (declaration) */
int main(void) {
    printf("%d\n", add(2, 3));      /* call */
    return 0;
}
int add(int a, int b) {         /* definition: header */
    return a + b;               /* body */
}
```

`return_type name(parameter list) { body; return value; }`. Parts: **prototype** (tells the compiler the types), **definition**, **call**. A function without a result has return type `void`. → 5

Q60 Differences between user-defined and library functions in C.

Asked in: 17th NTRCA · 17th NTRCA ICT (2023)

Library (built-in)	User-defined
already written, in the C library	written by the programmer
declared in a header: `#include <math.h>`	needs our own prototype and definition
name and job fixed	any name, any job
`printf()`, `scanf()`, `sqrt()`, `strlen()`	`add()`, `isPrime()`, `factorial()`

Q61 Write a function that checks whether a year is a leap year.

Asked in: BUET M.Sc. May 2016

```c
/* int isLeap(int y) {
       return (y % 4 == 0 && y % 100 != 0) || y % 400 == 0; } */
printf("%d %d %d %d\n", isLeap(2024), isLeap(1900), isLeap(2000), isLeap(2023));
```

Divisible by 4, except centuries, unless divisible by 400: **1900 no, 2000 yes**. → 1 0 1 0

MCQ as asked	answer	asked in
`a = func3(6) - func2(4,5) / func1(1,2,3);` Which function is called first?	**(d) cannot be predicted**	BUET IICT April 2021
Python function (the stem was not preserved): `def add(a, b=0): return a + b` — def, default argument, return	—	Combined Bank Officer(IT) 2026

Precedence says which operation happens first, not which operand is evaluated first: C leaves the order of evaluating `func1`, `func2`, `func3` unspecified.

Memory trick Value = **photocopy** (scribble on it, the original is safe). Reference = **the original file** (every change sticks).

22 Recursion and the recursive factorial ★ most asked · 13 exams

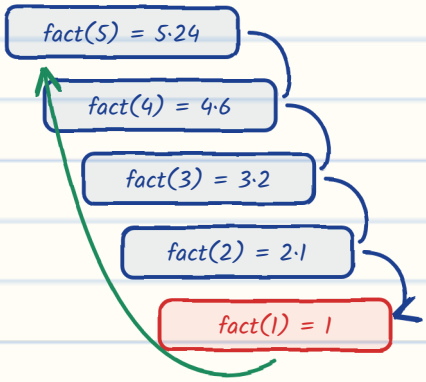
Q62 What do you mean by recursion? Calculate the factorial using recursion with C code. / Define recursion; write a recursive C factorial program. / Write a program for the factorial using the recursion method (C or Java). / Write a C program for the factorial of a positive integer.

✎ **Asked in:** 45th BCS ICT (2024) · BPSC ANE 2020 · BPSC AP 2020 · BPSC AP 2022 · Milk Vita AM(CSE) 2023 · BBA AME 2025 · ICT Ministry AP 2021 · 46th BCS ICT · 18th NTRCA College ICT (2024)

Recursion a function that **calls itself** on a smaller version of the problem. Every recursive function needs a **base case** (stops the calls) and a **recursive case** that moves toward it; without a base case the stack overflows.

```c
#include <stdio.h>
long long fact(int n) {
    if (n <= 1) return 1;          /* base case */
    return n * fact(n - 1);        /* recursive case */
}
int main(void) {
    int n;
    scanf("%d", &n);
    printf("%d! = %lld\n", n, fact(n));
    return 0;
}
```

input 5 → 5! = 120



results return upward → 120

Q63 What is a recursive function? Give an example. / What are recursion and a recursive function? Explain with an example. / Explain recursion and parameter passing with examples.

✎ **Asked in:** W2PDCL AE(CSE) 2019 · 15th NTRCA · 15th NTRCA ICT · 17th NTRCA ICT (2023)

Use the factorial above as the example, and draw the call stack. Parameter passing in it is **by value**: each call gets its own copy of n on the stack (call by value vs reference: previous sheets).

Q64 How do recursion and iteration differ in stack usage and memory overhead? Explain with examples.

✎ **Asked in:** 47th BCS CSE

	Recursion	Iteration (loop)
stack	one stack frame per call : fact(n) holds n frames → O(n) memory	one frame, a few variables → O(1)
overhead	call/return cost, risk of **stack overflow** for deep n	none beyond the loop
code	short, natural for trees, divide-and-conquer	longer for such problems

23 Fibonacci, GCD, digit sum, palindrome

12 exams

Q65 Print the Fibonacci series without recursion and with recursion. / Display the first ten Fibonacci numbers. / Find the n-th Fibonacci number using recursion. / Show the Fibonacci series from 0 up to n.

✎ **Asked in:** 17th NTRCA CSE (2023) · BB AP 2016 · 17th NTRCA ICT (2023) · 17th NTRCA · KGDCL AE(CSE) 2021 · NWPGCL JAM 2020

```c
#include <stdio.h>
int fib(int n) { return n < 2 ? n : fib(n - 1) + fib(n - 2); }
int main(void) {
    int a = 0, b = 1, c;
    for (int i = 0; i < 10; i++) {            /* loop */
        printf("%d ", a);  c = a + b;  a = b;  b = c;
    }
    printf("\nfib(9) = %d\n", fib(9));        /* recursion */
    return 0;
}
```

→ 0 1 1 2 3 5 8 13 21 34 · fib(9) = 34

The recursive version is exponential (fib(n−1) and fib(n−2) recompute the same values); the loop is O(n). "0 up to n": loop while a <= n.

Q66 Write a recursive C function for the GCD of two integers; using it, explain the steps for GCD(96, 36).

✎ **Asked in:** 47th BCS CSE

```c
/* int gcd(int a, int b) { return b == 0 ? a : gcd(b, a % b); } */
printf("%d\n", gcd(96, 36));
```

gcd(96, 36) → gcd(36, 24) → gcd(24, 12) → gcd(12, 0) → 12.

→ 12

Q67 Write a recursive function that returns the sum of the digits of a number (3426 → 15). / Write a function that separates each digit of an integer; you must use recursion.

✎ **Asked in:** GTCL AE(CSE) 2022 · Parliament Secretariat SAE 2023 · MoFood Network/Website Manager 2025 · BUET M.Sc. April 2017

```c
/* int sumd(int n) { return n == 0 ? 0 : n % 10 + sumd(n / 10); }
   void digits(int n) { if (n >= 10) digits(n / 10);
                        printf("%d ", n % 10); }   // print after the call */
printf("%d\n", sumd(3426));  digits(3426);
```

Printing **after** the recursive call gives the digits left to right. → 15 · 3 4 2 6

Q68 Write a recursive function that takes a string and returns a Boolean: is it a palindrome? / Write a recursive function for the scenario `for (i = 1; i < n; i++) for (j = 0; j < i; j++) for (k = 0; k < i; k++) x = x + 1;`

✎ **Asked in:** Combined Bank SO(IT) 2022 · Titas Gas AE(CSE) 2024

```c
/* int pal(const char *s, int i, int j) {
       if (i >= j) return 1;                  // met in the middle
       if (s[i] != s[j]) return 0;
       return pal(s, i + 1, j - 1); }
   int count(int n) { return n <= 1 ? 0 : count(n - 1) + (n - 1) * (n - 1); } */
printf("%d %d %d\n", pal("madam", 0, 4), pal("hello", 0, 4), count(4));
```

For each i the two inner loops add i × i, so x = Σ i² for i = 1 … n−1: count(n) = count(n−1) + (n−1)². → 1 0 14 (n = 4: 1 + 4 + 9).

24 Arrays: declare, pass, size, min/max, reverse

17 exams

Q69 Differentiate between the array and structure data types. / Write the difference between structure and array.

✎ **Asked in:** BOF AE(CSE) 2021 · BPSC AE 2022 · BPSC AP 2021 · BPSC AP 2022

Array	Structure
collection of elements of the same type	collection of members of different types
accessed by **index**: `a[2]`	accessed by **name**: `s.roll`
derived type, no keyword	user-defined, keyword `struct`
array name = address of element 0	structure name is not a pointer
`int marks[5];`	`struct student { int roll; char name[20]; float cgpa; };`

Q70 How can we initialise a 1-D array? How can we pass an array to a function in C?

✎ **Asked in:** 45th BCS CSE (2024)

```c
#include <stdio.h>
int minOf(int a[], int n) {          /* also: int *a */
    int m = a[0];
    for (int i = 1; i < n; i++) if (a[i] < m) m = a[i];
    return m;
}
int main(void) {
    int a[] = {7, 3, 9, 1, 5};       /* size from the list */
    int b[5] = {1, 2};               /* rest become 0 */
    printf("min = %d, b[4] = %d\n", minOf(a, 5), b[4]);
    return 0;
}
```

Initialise at declaration (`int a[5] = {…}`, size may be omitted, missing values become 0) or in a loop. An array is passed as the address of its first element, so pass the length too; the function can change the caller's elements.

→ min = 1, b[4] = 0

- Passing an array to a function passes: **C. the address of the first element.** ✎ BPSC AME 2019
- Size of `int array[15]` with 4-byte int: 15 × 4 = **60.** ✎ BPSC AP 2022
- Address difference between the 1st and 4th `float` (4 bytes): 3 × 4 = **(a) 12.** ✎ BUET IICT April 2021
- Python **list** holds any mix of types and grows freely; an **array** (`array` module / NumPy) holds one type, uses less memory and supports fast element-wise maths. ✎ W2PDCL SAE 2023

Exam trap C never checks array bounds: for `int a[5]` the valid indexes are 0 … 4. `a[5]` reads past the end (undefined behaviour), and a loop written `i <= n` instead of `i < n` does exactly that.

25 Array programs: min/max, reverse, averages

12 exams

Q71 Write a function returning the minimum (smallest) value of an array. / Read 10 array elements and find the maximum. / Find the maximum price of 20 items. / Read N numbers and print the largest and the smallest.

✎ **Asked in:** BSCL AE(CSE) 2025 · Combined Bank AP 2023 · BBA AME 2025 · BB AP 2019 · BKSP AP 2022

```
int n, a[100], max, min;
scanf("%d", &n);
for (int i = 0; i < n; i++) scanf("%d", &a[i]);
max = min = a[0];
for (int i = 1; i < n; i++) {
    if (a[i] > max) max = a[i];
    if (a[i] < min) min = a[i];
}
printf("max = %d, min = %d\n", max, min);
```

Same loop for 10 elements or 20 prices: set n.

5 numbers 40 15 90 25 60 → max = 90, min = 15

Q72 Write a C program that stores some data in an array and prints them in reverse order.

✎ **Asked in:** 17th NTRCA ICT (2023) · 17th NTRCA

```
int n, a[100];
scanf("%d", &n);
for (int i = 0; i < n; i++) scanf("%d", &a[i]);
for (int i = n - 1; i >= 0; i--) printf("%d ", a[i]);
```

10 20 30 40 50 → 50 40 30 20 10

Q73 Read five numbers into an array and find their average. / Find the sum, product and average of numbers in an array. / N students' marks: find the average and list who is above and below it.

✎ **Asked in:** 16th NTRCA ICT · 16th NTRCA · BOF AP 2022

```
int n;  double m[100], sum = 0, prod = 1, avg;
scanf("%d", &n);
for (int i = 0; i < n; i++) { scanf("%lf", &m[i]); sum += m[i]; prod *= m[i]; }
avg = sum / n;
printf("sum %.0f, product %.0f, average %.2f\n", sum, prod, avg);
for (int i = 0; i < n; i++)
    printf("student %d: %s\n", i + 1, m[i] >= avg ? "above" : "below");
```

70 45 88 60 52 → sum 315, product 864864000, average 63.00 · 1 above, 2 below, 3 above, 4 below, 5 below

Q74 A function receives an integer array and prints the numbers divisible by 3. / Read 10 integers and find the average of those divisible by 5.

✎ **Asked in:** Combined Bank Officer(IT) 2024 · DESCO SAE 2022

```
/* void div3(int a[], int n) { for (...) if (a[i] % 3 == 0) printf("%d ", a[i]); } */
int a[10] = {3, 10, 12, 25, 7, 30, 18, 5, 40, 9}, cnt = 0, sum = 0;
div3(a, 10);
for (int i = 0; i < 10; i++) if (a[i] % 5 == 0) { sum += a[i]; cnt++; }
if (cnt) printf("average of multiples of 5 = %.2f\n", (double) sum / cnt);
```

→ 3 12 30 18 9 · average of multiples of 5 = 22.00 (10, 25, 30, 5, 40)

26 Missing number, common elements, Kadane

6 exams

Q75 An array of size 10 holds 1 to 10 in random order, but one number was replaced by 0. Write a C/C++ program (with functions) to restore the lost number.

✎ **Asked in:** BAERA Engineer(CSE) 2023 · MoFood Network/Website Manager 2025

```
/* int restore(int a[], int n): s = sum of a; zero at index z;
      a[z] = n(n+1)/2 - s;  return a[z]; */
int a[10] = {4, 9, 1, 0, 7, 2, 10, 3, 8, 5};
printf("missing = %d\n", restore(a, 10));
```

1 + … + 10 = 55, the array sums to 49, so the zero was 6. One pass, O(n), no sorting. → missing = 6

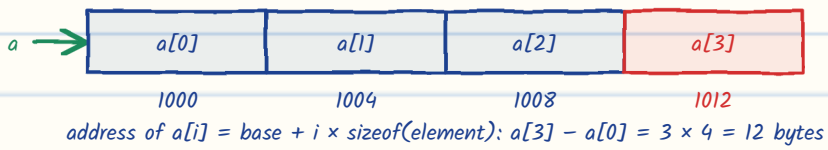
Q76 Print the common elements of two arrays and their count (elements distinct). / Report whether an integer array is ascending, descending or unordered. / Maximum subarray: find the contiguous subarray with the largest sum in [−2, 1, −3, −1, 2, 1, −5, 4]. / Array {4,5,6,7}: print every pair {4,5},{4,6},…,{6,7}.

✎ **Asked in:** BB AD(ICT) 2025 · BUET M.Sc. · BPDB AE(CSE) 2023 · NWPGCL AE(CSE)

```
int A[] = {1, 4, 6, 9}, B[] = {2, 4, 9, 11}, c = 0;
for (int i = 0; i < 4; i++) for (int j = 0; j < 4; j++)
    if (A[i] == B[j]) { printf("%d ", A[i]); c++; }
printf("(count %d)\n", c);
int s[] = {2, 1, 0, -1, -2, -3}, up = 1, down = 1;
for (int i = 1; i < 6; i++) { if (s[i] <= s[i-1]) up = 0; if (s[i] >= s[i-1]) down = 0; }
printf("%s\n", up ? "ascending" : down ? "descending" : "unordered");
int m[] = {-2, 1, -3, -1, 2, 1, -5, 4}, best = m[0], cur = m[0];
for (int i = 1; i < 8; i++) {                    /* Kadane */
    cur = (cur + m[i] > m[i]) ? cur + m[i] : m[i];
    if (cur > best) best = cur;
}
printf("max subarray sum = %d\n", best);
int p[] = {4, 5, 6, 7};
for (int i = 0; i < 4; i++) for (int j = i + 1; j < 4; j++) printf("{%d,%d} ", p[i], p[j]);
```

→ 4 9 (count 2) · descending · max subarray sum = 4 · {4,5} {4,6} {4,7} {5,6} {5,7} {6,7}

Kadane's algorithm (O(n)): keep the best sum ending here (extend or restart) and the best seen. Here the answer is the single element **[4]**; [2, 1] gives only 3.



address of a[i] = base + i × sizeof(element): a[3] − a[0] = 3 × 4 = 12 bytes

an array is one block; its name is the address of a[0]

27 Matrix programs, sorting and searching

4 exams

Q77 Take matrices A and B as input (handle different dimensions): (a) C = A × B (b) the average of A and of B (c) the maximum of C.

✎ **Asked in:** Combined Bank SO(IT) 2023

```
int r1, c1, r2, c2;  double A[10][10], B[10][10], C[10][10], sa = 0, sb = 0, mx;
scanf("%d %d", &r1, &c1);
for (int i = 0; i < r1; i++) for (int j = 0; j < c1; j++) { scanf("%lf", &A[i][j]); sa += A[i][j]; }
scanf("%d %d", &r2, &c2);
for (int i = 0; i < r2; i++) for (int j = 0; j < c2; j++) { scanf("%lf", &B[i][j]); sb += B[i][j]; }
if (c1 != r2) { printf("Cannot multiply: columns of A != rows of B\n"); return 0; }
for (int i = 0; i < r1; i++) for (int j = 0; j < c2; j++) {
    C[i][j] = 0;
    for (int k = 0; k < c1; k++) C[i][j] += A[i][k] * B[k][j];
}
mx = C[0][0];
for (int i = 0; i < r1; i++) { for (int j = 0; j < c2; j++) { printf("%6.0f", C[i][j]); if (C[i][j] > mx) mx = C[i][j]; } print
printf("avg A %.2f, avg B %.2f, max C %.0f\n", sa / (r1 * c1), sb / (r2 * c2), mx);
```

A (2×3) = 1 2 3 / 4 5 6, B (3×2) = 7 8 / 9 10 / 11 12 → C = 58 64 / 139 154 · avg A 3.50, avg B 9.50, max C 154

Q78 Read an m × n matrix (m, n from the user) and print each row followed by its sum, then the column sums (sample on the right).

✎ **Asked in:** Combined Bank SO(IT) 2024

```
int m, n, a[10][10], col[10] = {0};
scanf("%d %d", &m, &n);
for (int i = 0; i < m; i++) {
    int row = 0;
    for (int j = 0; j < n; j++) { scanf("%d", &a[i][j]); row += a[i][j]; col[j] += a[i][j]; printf("%d ", a[i][j]); }
    printf("%d\n", row);
}
for (int j = 0; j < n; j++) printf("%d ", col[j]);
```

→ 1
4 2
→ 2
5 3
→ 3
2 1 8
→ 6
11 6
(colu
sums

Q79 Write a C program to sort in ascending order using selection sort. / Write a C/C++ program for binary search.

✎ **Asked in:** BCC AP 2017 · Petrobangla AM(IT) 2022

```
int a[] = {64, 25, 12, 22, 11}, n = 5, key = 22, lo = 0, hi = 4, mid, t;
for (int i = 0; i < n - 1; i++) {                  /* selection sort */
    int m = i;
    for (int j = i + 1; j < n; j++) if (a[j] < a[m]) m = j;
    t = a[i]; a[i] = a[m]; a[m] = t;               /* smallest to front */
}
for (int i = 0; i < n; i++) printf("%d ", a[i]);
while (lo <= hi) {                                 /* binary search */
    mid = (lo + hi) / 2;
    if (a[mid] == key) { printf("\nfound %d at index %d\n", key, mid); break; }
    if (a[mid] < key) lo = mid + 1; else hi = mid - 1;
}
```

→ 11 12 22 25 64
→ found 22 at index 2

Selection sort: n − 1 passes, smallest to the front: $O(n^2)$. Binary search needs a **sorted** array, halves the range: $O(\log n)$.

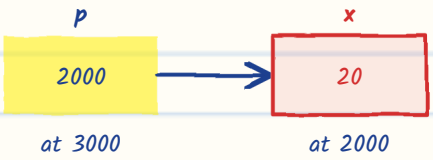
28 Pointers, malloc vs calloc, DMA

12 exams

Q80 What is a pointer? Write the advantages (and disadvantages) of using pointers. / Why are pointers needed? How is a pointer declared?

✎ **Asked in:** 17th NTRCA CSE (2023) · 17th NTRCA ICT (2023) · 17th NTRCA · 15th NTRCA CSE · 15th NTRCA ICT

Pointer a variable that stores the memory address of another variable. Declare `type *name;` · `&x` = address of `x` · `*p` = value at the address (dereference).




```
int x = 10;
int *p = &x;          /* p holds x's address */
*p = 20;              /* changes x through p */
printf("x = %d, *p = %d\n", x, *p);
```

p stores the address of x

→ x = 20, *p = 20

Advantages

- ✓ call by reference: a function can change the caller's data
- ✓ dynamic memory (malloc) and data structures (linked list, tree)
- ✓ fast array and string handling
- ✓ return several values; pass big structs cheaply

Disadvantages

- ✓ uninitialised / dangling pointers crash the program
- ✓ memory leaks if `free` is forgotten
- ✓ hard to read and debug
- ✓ can overwrite any memory

Q81 What is a pointer? Write the differences between malloc() and calloc(). / Describe dynamic memory allocation in C. / How does dynamic memory allocation help in managing data?

✎ **Asked in:** 15th NTRCA ICT · 17th NTRCA ICT (2023) · 15th NTRCA · SPCBL SAE 2022 · 37th BCS

	`malloc(size)`	`calloc(n, size)`
arguments	one: total bytes	two: count and size of each
initial value	garbage	all bytes 0
speed	faster	a little slower (zeroing)

```
int n = 5, *a = calloc(n, sizeof(int));   /* zeroed */
if (a == NULL) return 1;                  /* check! */
for (int i = 0; i < n; i++) a[i] = i * i;
a = realloc(a, 2 * n * sizeof(int));      /* grow */
printf("%d %d\n", a[4], n * 2);
free(a);                                  /* give back */
```

Dynamic memory allocation reserves memory on the **heap at run time** (`malloc`, `calloc`, `realloc`) and returns it with `free`, so a program uses exactly as much as the data needs (unknown n, growing lists) instead of a fixed array. → 16 10

29 Pointer arithmetic and reading declarations

8 exams

Q82 How does pointer arithmetic behave in C? Illustrate with an example. / Five short questions on pointer arithmetic, including multidimensional arrays.

✎ **Asked in:** 47th BCS CSE · BUET M.Sc. October 2017

```
int a[5] = {10, 20, 30, 40, 50}, *p = a;
int m[2][3] = {{1, 2, 3}, {4, 5, 6}};
printf("%d %d %d\n", *(p + 2), p[4], *p + 1);     /* 30 50 11 */
printf("%ld\n", (long) (&a[4] - &a[0]));          /* elements apart: 4 */
printf("%d %d\n", *(*(m + 1) + 2), *(m[0] + 4));  /* m[1][2], m[1][1] */
```

`p + 1` moves by `sizeof(*p)` bytes, not 1 byte; subtracting two pointers gives the number of **elements** between them. `*(*(m + i) + j) = m[i][j];` rows are stored one after another, so `m[0] + 4` lands on `m[1][1]`. → 30 50 11 · 4 · 6 5

MCQ as asked	answer	asked in
Declare an array Arr of four pointers to float	`float *Arr[4];`	49th BCS ICT (2025)
Function pointer f to a function taking (int, float) and returning double	`double (*f)(int, float)`	49th BCS ICT (2025)
`int (*a)[10];` declares …	**a pointer to an array of 10 ints**	BPSC AME 2019
Pass a function pointer: (a) `void pass(int (*fptr)(int, float, char)){}` (b) `void pass(*fptr(…)){}` …	**(a)**	BUET IICT April 2021
p → float array at 0x002a32e4; value of `&p[5]`?	**(c) 0x002a32f8**	BUET M.Sc. April 2021
Pair NOT treated as identical: `foo(int *i)`/`foo(int i[])` · `a[i]`/`i[a]` · `char c[10]`/`char *c` · `bar(int)`/`bar(int x)`	`char c[10];` `char *c;`	BPSC AME 2022

```
float f[6];  float *p = f;
printf("%ld bytes\n", (long) ((char *) &p[5] - (char *) p));   /* 5 * 4 = 20 = 0x14 */
int a[3] = {7, 8, 9};  printf("%d %d\n", a[2], 2[a]);          /* a[i] == *(a+i) == i[a] */
```

→ 20 bytes (0x002a32e4 + 0x14 = 0x002a32f8) · 9 9

Memory trick Read a declaration **from the name outward, right before left**: `int (*a)[10]` → *a is a pointer to an array of 10 ints*; `float *Arr[4]` → *Arr is an array of 4 pointers to float*.

Exam trap `a[i] = i[a] = *(a + i);` a parameter `int i[]` is `int *i`. Only a **declared array** differs from a pointer.

30 Strings without library functions 8 exams

Q83 Write a program to reverse a string without using a library function. / Write a function to reverse a string without a global variable. / Write a C++ method to reverse a char array without a temp array or STL functions. / Find the length of a string without `strlen()`.

✎ **Asked in:** BREB AP 2023 · Combined Bank ADA 2022 · BUET M.Sc. May 2019 · BUET M.Sc. May 2016 · BPSC AME 2021

```
/* int len(const char *s) { int n = 0; while (s[n] != '\0') n++; return n; }
   void reverse(char *s) {                    // in place: two ends meet
       for (int i = 0, j = len(s) - 1; i < j; i++, j--)
           { char t = s[i]; s[i] = s[j]; s[j] = t; } }          */
char s[] = "Bangladesh";
printf("%d ", len(s));  reverse(s);  printf("%s\n", s);
```

Length: count characters up to the terminating `'\0'`. Reverse: swap the two ends and move inward — in place, so no second array, no global (only one `char` for the swap). → 10 hsedalgnaB

Q84 Write a Java program counting the vowels and consonants in a given string.

✎ **Asked in:** BOF AP 2022

```
String s = "Programming in Java".toLowerCase();
int v = 0, c = 0;
for (char ch : s.toCharArray())
    if (ch >= 'a' && ch <= 'z') { if ("aeiou".indexOf(ch) >= 0) v++; else c++; }
System.out.println("vowels " + v + ", consonants " + c);
```

Count only letters: spaces and digits are neither. → vowels 6, consonants 11

Q85 Write a C or Java function converting a string to an integer without built-in conversion functions. / Replace each letter with the next letter (case preserved, z → a, non-letters unchanged) and state the time complexity.

✎ **Asked in:** DPDC AM(ICT) 2025 · ICB Securities AP 2026

```
/* int toInt(const char *s): optional '-', then v = v*10 + (c - '0')
   void shift(char *s): letter → (c - 'a' + 1) % 26 + 'a', same for A-Z */
char w[] = "Zebra-9y";
printf("%d %d\n", toInt("-4512") + 2, toInt("307"));
shift(w);  printf("%s\n", w);
```

Both are one pass over the string: **O(n)** time, O(1) extra space. → -4510 307 · Afcsb-9z

Q86 Write a C program that takes integers as command-line arguments and prints their sum.

✎ **Asked in:** DPDC AM(ICT) 2025

```
#include <stdio.h>
#include <stdlib.h>
int main(int argc, char *argv[]) {          /* argv[0] = program name */
    long sum = 0;
    for (int i = 1; i < argc; i++) sum += atol(argv[i]);
    printf("sum = %ld\n", sum);
    return 0;
}
```

Run `./sum 10 20 -5 7`: `argc` = 5, the numbers arrive as strings, converted with `atol`. → sum = 32

- **Array vs pointer**: an array is a fixed block (`sizeof a` = whole array, its name cannot be reassigned); a pointer is one variable holding an address (`sizeof p` = 8, can point anywhere, can be changed: `p++`).
  ✎ BPSC AME 2021

31 Remove a substring, IP class, IP validation

4 exams

Q87 Write a C program to remove a given substring from a string, e.g. input "programming", remove "gram" → output "proming".

✎ **Asked in:** RPGCL AM(ICT) 2022

```
char s[100] = "programming", sub[] = "gram";
char *p;
int k = strlen(sub);
while ((p = strstr(s, sub)) != NULL)          /* find it */
    memmove(p, p + k, strlen(p + k) + 1);     /* close the gap */
printf("%s\n", s);
```

→ proming

Exam trap The printed solution deletes every copy of one **character**, not the substring, so it cannot give "proming"; it also uses `conio.h` and `void main`.

Q88 Given an IPv4 address string, write C/C++/Java code to show the class the address belongs to (sample 192.168.0.1). / Identify the class of an IP address.

✎ **Asked in:** Rupali Bank ANE 2021 · JBL ANA 2022

```
/* char ipclass(const char *ip) {
       int a = atoi(ip);                  // first octet
       return a < 128 ? 'A' : a < 192 ? 'B' : a < 224 ? 'C' : a < 240 ? 'D' : 'E'; } */
printf("%c %c %c\n", ipclass("192.168.0.1"), ipclass("10.0.0.5"), ipclass("172.16.4.2"));
```

class	first octet
A	0–127
B	128–191
C	192–223
D	224–239 (multicast)
E	240–255

→ C A B

Q89 Write a program that validates IPv4 addresses and prints "valid" or "not valid" (e.g. 128.0.0.1 and 125.16.100.1 valid; 125.512.100.1 and 125.512.100.abc not valid).

✎ **Asked in:** RPGCL AM(ICT) 2022

```
/* int valid(const char *ip): for each part: 1–3 digits, no leading 0, value ≤ 255;
       parts separated by '.', exactly 4 parts, nothing else */
const char *t[] = {"128.0.0.1", "125.16.100.1", "125.512.100.1", "125.512.100.abc", "1.2.3"};
for (int i = 0; i < 5; i++) printf("%s: %s\n", t[i], valid(t[i]) ? "valid" : "not valid");
```

→ 128.0.0.1 valid · 125.16.100.1 valid · 125.512.100.1 not valid · 125.512.100.abc not valid · 1.2.3 not valid

Rules to state: four parts separated by dots; each part only digits, 0–255, no leading zero ("01" is invalid); no empty part. Parse by hand (or `strtok` on ".") and check every rule.

32 Structure vs union, nested structure, structure + malloc

7 exams

Q90 What are structure and union? Write their differences with examples. / Explain the differences between structure and union with examples. / Describe structure and union and how to declare them. / Write the rules for declaring a union.

Asked in: 16th NTRCA · 16th NTRCA ICT · 47th BCS CSE · 15th NTRCA CSE · 13th NTRCA CSE (2016)

	Structure (struct)	Union (union)
memory	each member has its own space	all members share one space
size	sum of members (+ padding)	size of the largest member
values at once	all members	only the last one written
use	a record (student, employee)	save memory; one of several types

```c
#include <stdio.h>
struct S { int i; char c; float f; };    /* struct tag { members }; */
union  U { int i; char c; float f; };    /* same syntax, keyword union */
int main(void) {
    union U u;
    u.i = 65;                            /* write int, read char */
    printf("%zu %zu %c\n", sizeof(struct S), sizeof(union U), u.c);
    return 0;
}
```

Declaring a union: `union tag { type m1; type m2; … };` then `union tag var;`, members reached with `.` (or `->` through a pointer); only one member is valid at a time. → 12 4 A (4 + 1 + 3 padding + 4).

Q91 What is a nested structure in C? Explain with an example.

Asked in: SPCBL SAE 2022

```c
struct date { int d, m, y; };
struct emp { char name[20]; struct date join; };   /* struct inside struct */
struct emp e = {"Rina", {1, 7, 2024}};
printf("%s joined %d/%d/%d\n", e.name, e.join.d, e.join.m, e.join.y);
```

A nested structure has a structure as a member; reach inner members with two dots. → Rina joined 1/7/2024

Q92 Write a program to store information using structures with dynamic memory allocation.

Asked in: 38th BCS (2018)

```c
struct course { int marks; char subject[30]; };
int n;
scanf("%d", &n);
struct course *p = malloc(n * sizeof(struct course));
if (p == NULL) return 1;
for (int i = 0; i < n; i++) scanf("%d %29s", &p[i].marks, p[i].subject);
for (int i = 0; i < n; i++) printf("%s: %d\n", (p + i)->subject, (p + i)->marks);
free(p);
```

input 2 · 75 Physics · 82 Chemistry →
Physics: 75
Chemistry: 82

33 getc vs getch, formatted I/O, files, argv

12 exams

Q93 Differences between getc(), getchar(), getch() and getche(). / Difference between getch() and getche().

✎ **Asked in:** 45th BCS ICT (2024) · BBL AM(ICT) 2019

function	reads from	echo on screen?	waits for Enter?	header
`getc(fp)`	any stream / file	—	buffered	stdio.h
`getchar()`	stdin (= `getc(stdin)`)	yes	yes	stdio.h
`getch()`	keyboard	**no**	no	conio.h (non-standard)
`getche()`	keyboard	**yes** (e = echo)	no	conio.h

- **File pointer**: `FILE *fp = fopen("data.txt", "r");` — a pointer to a `FILE` structure that keeps the file's position and buffer; used by `fscanf`, `fprintf`, `fgets`, `fclose`. ✎ BBL AM(ICT) 2019
- **Formatted I/O** = `printf` / `scanf` with format specifiers that convert between text and values. Keyboard input: `scanf("%d", &x);`. `%d` means a signed decimal `int` (others: `%f` `%c` `%s` `%lf`). ✎ BPSC ANE 2020

Q94 myfile.txt has lines "day, demand, shortage" (1, 43571, 443 / 2, 34554, 432 / ...). Write a C program to find the power shortage of the 30th day.

✎ **Asked in:** BPDB AE(CSE) 2018

```
FILE *fp = fopen("myfile.txt", "r");
int day, demand, shortage;
if (fp == NULL) { printf("cannot open\n"); return 1; }
while (fscanf(fp, "%d, %d, %d", &day, &demand, &shortage) == 3)
    if (day == 30) { printf("Day 30 shortage = %d\n", shortage); break; }
fclose(fp);
```

sample file with day 30 = 517
→ Day 30 shortage = 517

Q95 Write a Python function that takes a filename and prints its first 10 lines.

✎ **Asked in:** BCC AP 2021

```
from itertools import islice
def head(filename, n=10):
    with open(filename) as f:
        for line in islice(f, n):
            print(line, end="")
head("notes.txt")
```

12-line file → prints 11 ... 110

Exam trap The printed answer is Python 2 (`xrange`, `print head`), not a function, and crashes on a file shorter than 10 lines; `islice` stops safely.

34 Student records from a file: totals, grades, toppers

ICB 2026 + 3 unrecalled

Q96 student.txt holds records: StudentID Name Marks1 Marks2 Marks3 Marks4. Write a program that (a) 3 calculates total and average marks; (b) 3 assigns grades: ≥ 80 A+, ≥ 70 A, ≥ 60 B, ≥ 50 C, < 50 F; (c) 3 displays the top three students; (d) 3 displays the subject with the highest average.

✎ **Asked in:** ICB Securities AP 2026(Aptitude)

```
struct st { int id; char name[20]; int m[4]; int total; double avg; } s[50], t;
int n = 0;  double sub[4] = {0};
FILE *fp = fopen("student.txt", "r");
if (!fp) return 1;
while (n < 50 && fscanf(fp, "%d %19s %d %d %d %d", &s[n].id, s[n].name,
                        &s[n].m[0], &s[n].m[1], &s[n].m[2], &s[n].m[3]) == 6) {
    s[n].total = 0;
    for (int k = 0; k < 4; k++) { s[n].total += s[n].m[k]; sub[k] += s[n].m[k]; }
    s[n].avg = s[n].total / 4.0;
    n++;
}
fclose(fp);
for (int i = 0; i < n; i++) {                                   /* (a), (b) */
    double a = s[i].avg;
    const char *g = a >= 80 ? "A+" : a >= 70 ? "A" : a >= 60 ? "B" : a >= 50 ? "C" : "F";
    printf("%d %-6s total %d avg %.2f grade %s\n", s[i].id, s[i].name, s[i].total, a, g);
}
for (int i = 0; i < n - 1; i++)                                 /* (c) sort by total */
    for (int j = 0; j < n - 1 - i; j++)
        if (s[j].total < s[j + 1].total) { t = s[j]; s[j] = s[j + 1]; s[j + 1] = t; }
printf("Top 3: %s, %s, %s\n", s[0].name, s[1].name, s[2].name);
int best = 0;                                                    /* (d) */
for (int k = 1; k < 4; k++) if (sub[k] > sub[best]) best = k;
printf("Highest-average subject: Marks%d (%.2f)\n", best + 1, sub[best] / n);
```

→ 101 Asif total 325 avg 81.25 grade A+ · 102 Mitu 254 63.50 B · 103 Rony 354 88.50 A+ · 104 Tania 185 46.25 F · 105 Sami 295 73.75 A · Top 3: Rony, Asif, Sami · Highest-average subject: Marks2 (72.40)

Marks are given for each part: read with `fscanf` until it fails (stop on a bad line), keep subject totals while reading, bubble-sort a copy by total for the toppers, and grade on the **average**.

Q97 A Python program / a program using a Teletalk number 015… / "C programme output determination" — the tasks were not recalled in any source.

✎ **Asked in:** PGCB SAE(Computer) 2020 · Teletalk Executive(Technical) 2015 · NHA AP 2019

Prepare with the general programs on these sheets (loops, digits, strings) and the output drills that follow; for a phone-number task, treat it as a digit/string problem (validate 11 digits starting with 015, count or sum the digits).

35 Output questions: if, logic, shifts, division

★ output tracing: 43 exams

Every output below was produced by compiling the exam's code (typos fixed as noted). Method: write the variables in a row, update them statement by statement, and remember **short-circuit**, **integer division** and **unsigned conversion**.

Code (as asked)	Output	Why	Asked in
`a = 85; if(a>=90) printf("A"); if(a>=80) printf("B"); if(a>=70) printf("C"); if(a>=60) printf("D"); else printf("F");`	**BCD**	four separate `if`s; the `else` belongs to the last one only	Petrobangla AM(CSE) 2024
`unsigned a = 3, b = 9; printf("a << 1 = %d ", a<<1); printf("b << 1 = %d", b<<1);`	**a << 1 = 6 b << 1 = 18**	left shift by 1 = × 2	WZPGCL AE(CSE) 2023
`a = -1, b = -a; x = (a>0)&&(b<0) || (a<0&&(b>0)); y = (a<=0) || (b>=0)&& (a>=0 || (b<=0)); printf("%d", x==y);`	**1**	`x = 0 || 1 = 1`; `y = 1` (short-circuit). Read `a>==0` as `a>=0`, `%/d` as `%d`	DMTCL AE(ICT) 2023
`i=-3, j=1, k=0; m = ++i || ++j && ++k; printf("%d,%d,%d, %d", i,j,k,m);`	**-2,1,0, 1**	`++i = −2` is true, so `||` skips the right side: j, k unchanged	DPDC JAM(ICT) 2020
`i=-20, j=-1, k=2; m = ++i && ++j && ++k;` *print i, j, k, m*	**-19, 0, 2, 0**	`++j = 0` is false: `&&` stops, k stays 2	BCC AP 2017
`a=0, b=1, c=2; *((a) ? &b : &a) = a ? b : c;` *print a, b, c*	**2, 1, 2**	a = 0 picks `&a` on the left and c on the right: a = 2	BCC AP 2017
`unsigned int i = 10; if (i > -1) cout << "Yes"; else cout << "No";`	**No**	−1 is converted to unsigned = 4294967295, so 10 > it is false	BUET M.Sc. April 2024
`x=3, y=2; if (x==3) y=2; else y=3; cout << x << " " << y;`	**(b) 3 2**	x == 3, so y = 2	BB AD(ICT) 2025
`calc(x)`: ≥50 → 50, ≤20 → 20, else 35; returns 2x. `y=25; z = calc(y);` *print y, z*	**(c) y = 25, z = 70**	y is passed by value; 25 → 35 → 70	BUET M.Sc. April 2021
`i=5; l = i / -2; k = i % -2; printf("%d %d", l, k); x = 5 % 1;`	**(a) -2 1** then 0	division truncates toward 0; the remainder takes the sign of the dividend	BUET IICT April 2021
pseudocode: a=15, b=7; a = a mod (a−3); b = b mod (b−3); a = a mod 1; b = b mod 1; print a+b	**(c) 0**	15 mod 12 = 3, 7 mod 4 = 3; anything mod 1 = 0	BUET IICT April 2021
`printf("%c", 100);`	**(c) d**	ASCII 100 = 'd' ('a' = 97)	DoICT ANE 2019

Memory trick **&& stops at the first false, || stops at the first true**: cross out everything after the deciding operand before you trace.

Exam trap Petrobangla asked the `a = 85` question three times (two copies lost the code): the answer is **BCD**, not "B", because the `if`s are not chained with `else`.

36 Increment side effects, macros, static, XOR

8 exams

Code (as asked)	Output	Why	Asked in
`a = 5, b = 18, c = 27; a = b++ + ++c; b = a++ - c--; c = a++ + b--;` values of a, b, c?	**a = 48, b = 17, c = 65**	a = 18 + 28 = 46 (b 19, c 28); b = 46 − 28 = 18 (a 47, c 27); c = 47 + 18 = 65 (a 48, b 17)	46th BCS CSE
`#define MAX(A,B) ((A)>(B)) ? A:B` · `i=1, j=2; val = MAX(++i, ++j);`	**4**	expands to `((++i)>(++j)) ? ++i : ++j`; 2 > 3 false, then ++j again → 4	DESCO AE(CSE) 2016
`int sum(int v){ static int r = 0; r += v; return r; }` · `for(i=1;i<10;i++) value = sum(i);`	**45**	`static` keeps r between calls: 1 + 2 + … + 9	DESCO AE(CSE) 2016
`register static int i = 10; i = 11; printf("%d", i);`	**(b) compile-time error**	a variable can have only one storage class	BUET IICT April 2021
`a=10, b=20; a ^= b; b ^= a; a ^= b; printf("%d %d", a, b);`	**(c) 20 10**	XOR swap	Combined Bank AHE/AE(IT) 2021
`a=10, b=25; a = b++ + a++; b = ++b + ++a; printf("%d%d n", a, b);`	**3663 n** (expected)	a = 25 + 10 = 35, b = 26; then b = 27 + 36 = 63. `a = … a++` is **undefined**; clang prints 3663 n. The `n` is printed literally (the `\` is missing)	DESCO AE(IT) 2022
`a = 1; printf("%d%d%d", ++a, a, a++);`	**undefined**	a is changed twice with no sequence point between arguments: clang prints **222**; the key (gcc, right to left) expects **331**	NWPGCL JAM(ICT) 2022
`i=4; printf("%d %d", ++i, i++); printf("%d", i++);` (a) 4 5 6 (b) 5 7 8 (c) 6 4 6 (d) 4 5 7	**undefined**; key (c)	same fault: gcc evaluates right to left → "6 4" then 6 → **(c)**; clang prints "5 5" then 6	Combined Bank AHE/AE(IT) 2021

Sequence point a point where all earlier side effects are finished: the end of a full expression (`;`), `&&`, `||`, `?:`, the comma operator, and a function call (after its arguments). Changing a variable **twice** between two sequence points — or reading it while changing it — is **undefined behaviour**: the C standard allows any result.

Exam trap Bangladeshi keys for `printf("%d %d", ++i, i++)` are built on gcc's right-to-left order. In the written exam write: "**undefined behaviour** (argument order unspecified); with gcc the output is …", and give the key's value.

Memory trick A **macro is text copy-paste**: write the expansion out in full before counting the ++.

37 Loops: trace tables and endless loops

6 exams

Q98 `sum = 0; for (i = -1, p = 0; i <= 10; i = i + 2) { p = i * i; if (i > 5) break; sum = sum + (i + p); printf("i=%d, p=%d, sum=%d\n", i, p, sum); } printf("Outside Loop, i=%d, p=%d, sum=%d", i, p, sum);`

Asked in: BGDCL AE(CSE) 2021

i	p = i²	i > 5?	sum
−1	1	no	0
1	1	no	2
3	9	no	14
5	25	no	44
7	49	yes: break	44

i=-1, p=1, sum=0
i=1, p=1, sum=2
i=3, p=9, sum=14
i=5, p=25, sum=44
Outside Loop, i=7, p=49, sum=44

Code (as asked)	Output	Why	Asked in
(a) `n = 10; for(;;) printf("%d", n);` or (b) `i = 2, j = 2; while (i ? --i : j++) printf("%d", i);`	(a) **10101010…** forever · (b) **1**	(a) `for(;;)` has no condition. (b) i = 2 → `--i` = 1 true, print 1; then `--i` = 0 false: stop	PSB AP 2018
(a) `char str[120] = "Digital Bangladesh"; n = strlen(str); str[4] = '\0'; printf("%s", str);` or (b) `for(i=0;i<5;i++){ if(i==3) continue; printf("%d\n", i); }`	(a) **Digi** · (b) **0 1 2 4**	(a) '\0' at index 4 ends the string. (b) `continue` skips 3	PSB Programmer 2018
`#define N 7 · char str[] = "abpqx "; for(i=0;i<N-2;i++) if(i%2) printf("%c", str[i]++); else printf("%c", str[i]--);`	**abpqx**	postfix ++/-- print first, change after; i = 0 … 4 (Char read as char)	BTCL AE(CSE) 2017
(i) Java `Integer a = new Integer(33), b = new Integer(33); if (a == b) … println(a + b + 'a');` (ii) `i = 7; for(; i != 0; i--) printf("x=%d", i--);` (iii) `j = 1; while (j++ < 50) println(j + "");` (iv) `unsigned a = .1, b = 2; printf("%d", a << a + b);`	(i) **163** · (ii) **x=7x=5x=3x=1x=-1…** forever · (iii) **2 … 50** · (iv) **0**	(i) two objects: == false, but both branches print 33 + 33 + 97. (ii) i falls by 2 a pass and jumps over 0. (iii) 49 lines (add the missing)). (iv) a = 0	Sonali Bank ADA 2020

Q99 *Find the output and analyse the time and space complexity:* `n; first = 0; secnd = 1; if (n >= 1) cout << first; else if (n <= 2) cout << secnd; for (i = 2; i < n; i++) { next = first + secnd; cout << next; first = secnd; secnd = next; }`

Asked in: Rupali Bank ANA 2023

For n = 5 it prints 0 1 2 3 — Fibonacci with the **second 1 missing**: the `else if` never prints `secnd` when n ≥ 1 (fix: print 0 and 1 for n ≥ 2). Time **O(n)** (one loop), space **O(1)** (three variables). The printed `next + first + secnd;` is read as `next = first + secnd;`.

38 Pointers, arrays, recursion, strings

13 exams

Code (as asked)	Output	Why	Asked in
`int arr[5] = {1,2,3,4,5}; int *p = (arr + 2); printf("%d", p[-1]);`	**(ক) 2**	p points at arr[2]; p[−1] = arr[1]	49th BCS CSE (2025)
`int fun(int *p){ *p = *p + 10; return *p; } · x = 5; cout << fun(&x);`	**(a) 15**	changed through the pointer	BB AD(ICT) 2025
`int a[] = {5,3,2,1,4}; int b[] = {66,55,30,20,45}; printf("%d", b[a[2]]);`	**30**	a[2] = 2, b[2] = 30 (printed arr[a] read as a[])	BSCPLC AM(Engineering) 2024
`int k = 7; int *p = &k; int **m = &p; printf("%d%d%d", k, *p, **m);`	**(c) 7 7 7** (printed 777)	all three reach k	BUET IICT April 2021
`int a = 100; int *p = &a + 2; *p = 22; printf("%d", a);`	**undefined**; key (a) 100	writes outside a: a itself stays 100 if the program survives; our run crashed	Combined Bank AHE/AE(IT) 2021
`int (*fp)(int, int); fp = mul;` (mul takes 3 ints) `printf("…%d", fp(2, 3));`	**(b) the code is incorrect**	incompatible function-pointer type; clang refuses, gcc warns and the call is UB	BUET IICT April 2021
`bitwise(2147483649)` swaps bit i with bit 31−i (mask1 from the right, mask2 from the left)	**(d) 2147483649**	2147483649 = bits 31 and 0 set: its mirror image is itself	BUET M.Sc. April 2021
`int function(int x[], int i){ int s = x[i]; if(i>0) s += function(x, i-1); printf("%d", s); return s; } · function(y, 2),` y = {1,3,2,8}	**146**	deepest call prints first: 1, then 1+3 = 4, then 4+2 = 6	BCC AP 2021
`fun(i){ if(i<=0) return; printf("%d, ", i--); fun(--i); printf("%d, ", i); } · fun(5)`	**5, 3, 1, -1, 1, 3,**	each call prints i, recurses with i − 2, then prints its own i (now i − 2) on the way back	Combined Bank AP 2024
(i) `foo(x, y){ if(x==0) return y; return foo(x-1, x+y); } · foo(3, 4)` (ii) `foo(char *a){ if(*a && *a != ' '){ foo(a+1); putchar(*a); } } · "COVID-19"`	(i) **10** · (ii) **91-DIVOC**	(i) (3,4)→(2,7)→(1,9)→(0,10). (ii) print after the call = reverse	BUET AP 2025
Java: `static int x = 5; fun(n){ if(n<=1) return 1; x = x + 2; return fun(n-1) + x; } · fun(3)`	**19**	x = 7, then fun(2): x = 9, returns 1 + 9 = 10; back in fun(3): 10 + x (now 9) = 19	Combined Bank SO(IT) 2026
`string cars[4] = {"Volvo","BMW","Ford","Mazda"}; cout << cars[3]; cout << cars[2];`	**(c) MazdaFord**	no space or newline between them	BUET IICT April 2021
`string S = "Wasa Authority", s = S, R = S;` loop: even i → `S[i] = toupper, s[i] = tolower;` print s and R; `return 0;` inside the loop	**s: wasa Authority r: Wasa Authority**	the return ends main after the first pass (i = 0): only s[0] became 'w'	Dhaka WASA AME(Network) 2025

Memory trick Recursion output: **print before the call → top-down order; print after the call → reverse order.**

39 How to answer any output question

11 exams (code not preserved)

Q100 Explain the while-loop of the given program step by step for inputs 1221 and 3456 and write the output (the program was not printed; it is the reverse-and-compare palindrome loop).

✎ **Asked in:** BAPEX AGM(ICT) 2023

n	n % 10	rev
1221	1	1
122	2	12
12	2	122
1	1	1221

`while (n != 0) { rev = rev * 10 + n % 10; n /= 10; }` then compare rev with the original: 1221 → 1221, palindrome; 3456 → 6543, not a palindrome.

The papers below set an output question whose code was **not preserved** in any copy. They test the same traps as the four previous sheets; drill those.

- "After compilation and execution, what will be the output?" — see the `++i || ++j && ++k` family. ✎ DPDC JAM(ICT) 2020
- "Write down the output of this program." ✎ DESCO AE(IT) 2022
- "Program analysing and writing output." ✎ DESCO AE(CSE) 2016
- "Find the output of the following program." ✎ Sonali Bank Officer(IT) 2021
- "Find output" (code not on the page). ✎ NWPGCL AE(CSE)
- "C programming output problem" (incomplete question). ✎ Teletalk AM(IT) 2023
- "Given Python code, find its output with complexity": count the loops — one loop over n is $O(n)$, nested loops $O(n^2)$, halving $O(\log n)$. ✎ RAKUB ANSE 2023
- "String compare code in C: write the output": `strcmp(a, b)` returns **0** if equal, a **negative** value if a comes first, **positive** if after; `a == b` compares addresses, not text. ✎ BUET M.Sc. October 2018
- "Will it compile? If so, what is the output?": check declarations, types, missing `;`, storage-class clashes (`register static`), pointer types, then trace. ✎ BUET M.Sc. October 2018
- "C code with pointers: is there a problem? If not, the output": look for an uninitialised pointer, returning the address of a local, out-of-bounds index, missing `free`, double `free`. ✎ BUET M.Sc. May 2019

→ The ten traps examiners set

1. **Short-circuit**: `&&` / `||` skip the right side (DPDC, BCC).
2. **Separate ifs** vs if-else chain (Petrobangla BCD).
3. **Integer division**, truncation toward 0, sign of % (BUET IICT).
4. **Unsigned vs −1** comparison (BUET 2024).
5. **Pre vs post ++**, and **undefined behaviour** when a variable changes twice in one expression.
6. **Macro expansion** evaluates arguments again (DESCO MAX).
7. **static** locals keep their value (DESCO sum = 45).
8. **Loop that skips its exit** (`i -= 2` past 0; `for(;;)`).
9. **'\0' cuts a string**; `%c` of a number prints its ASCII character.
10. **Recursion print order**: before the call vs after it.

★ Night before the exam

read in 10 minutes

1. **Call by value** = copy, caller unchanged · **by reference** = address (`int *x`, `f(&a)`), caller changes (5 exams).
2. **Recursion** = base case + recursive case; factorial `n * fact(n-1)`, Fibonacci, GCD `gcd(b, a % b)`; one stack frame per call.
3. **Compiler** whole program, fast, object file · **interpreter** line by line, stops at the first error · **linker** joins libraries · **loader** loads to RAM.
4. **Array vs structure**: same type by index vs mixed types by name · **struct vs union**: own space vs shared (size of the largest).
5. **malloc** garbage, one argument · **calloc** zeroes, two arguments · always `free`.
6. **while** tests first (0 or more) · **do-while** exit-controlled (at least once) · **break** leaves, **continue** skips.
7. **Digits**: % 10 peels, / 10 drops, × 10 + pushes · **LCM** = a × b / GCD · primes to $\sqrt{n}$.
8. **Output traps**: short-circuit, unsigned −1, integer division, macro re-evaluation, static, UB with ++.

→ ✓ Practice MCQs (all asked; answers at the bottom)

1. Exit-controlled loop: (a) for (b) while (c) do-while (d) none ✎ 49th BCS CSE (2025)

2. Keep a variable's value between calls: (a) auto (b) extern (c) static (d) register ✎ 49th BCS CSE (2025)

3. Array of four pointers to float: (a) `float Arr[4]` (b) `(float) Arr[4]` (c) `float *Arr[4]` (d) `Ptr (float) Arr[4]` ✎ 49th BCS ICT (2025)

4. Size of `int array[15]` (4-byte int): (a) 15 (b) 19 (c) 11 (d) 60 ✎ BPSC AP 2022

5. Passing an array passes: (a) all elements (b) the first element (c) the address of the first element (d) the last address ✎ BPSC AME 2019

6. `(x < 0) ? 10 : 15`, x = 5: (a) 10 (b) 15 (c) 0 (d) 1 ✎ NWPGCL JAM(ICT) 2022

7. Converts a whole program to machine language: (a) interpreter (b) converter (c) simulator (d) compiler ✎ BPSC AP 2022

8. `printf("%c", 100);` (a) 100 (b) one hundred (c) d (d) 0 ✎ DoICT ANE 2019

Answers: 1 c · 2 c · 3 c · 4 d · 5 c · 6 b · 7 d · 8 c

Memory trick Every C answer: **definition → table → program → sample output**. Programs without an output line lose marks.

Exam trap Write `int main(void)` and `return 0;`, not `void main` and `conio.h`: several printed solutions lose marks for that.